

(3)

Date: (19.08.2023)

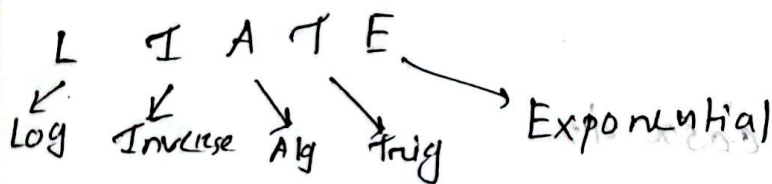
Class → 20

Date: / /

✓ Ch-7.2

✓ Integration by Parts:

$$\int uv \, dx = u \int v \, dx - \int \left\{ \frac{d}{dx} (u) \int v \, dx \right\} dx$$

Where, $u = f(x)$ $v = g(x)$ Example → (1-8)Problem: (Example-1)

$$\int x \cos x \, dx$$

$$= x \int \cos x \, dx - \int \left\{ \frac{d}{dx} (x) \int \cos x \, dx \right\} dx$$

$$= x \sin x - \int \sin x \, dx$$

$$= x \sin x + \cos x + C$$

[Ans.]

$$\int \textcircled{1} \ln x \, dx$$

Date : / /

Example $\rightarrow$ 3) $\int \ln x \, dx$

$$= \ln x \int dx - \int \left\{ \frac{d}{dx} (\ln x) \int dx \right\} dx$$

$$= x \ln x - \int \frac{1}{x} \cdot x \, dx$$

$$= x \ln x - \int dx$$

$$= x \ln x - x + c$$

[Ans.]

Example $\rightarrow$ 5) $I = \int e^x \cos x \, dx$

$$= \cos x \int e^x \, dx - \int \left\{ \frac{d}{dx} (\cos x) \int e^x \, dx \right\} dx$$

$$= e^x \cos x - \int -\sin x \cdot e^x \, dx$$

$$= e^x \cos x + \int e^x \sin x \, dx$$

$$= e^x \cos x + \left[\sin x \int e^x \, dx - \int \left\{ \frac{d}{dx} (\sin x) \int e^x \, dx \right\} dx \right]$$

$$= e^x \cos x + \left[e^x \sin x - \int \cos x \cdot e^x \, dx \right]$$

$$= e^x \cos x + \left[e^x \sin x - \int e^x \cos x \, dx \right]$$

$$= e^x \cos x + e^x \sin x - I$$

$$\Rightarrow 2I = e^x (\sin x + \cos x) \quad (\text{PO - sign})$$

$$\therefore I = \frac{1}{2} e^x (\cos x + \sin x) + C$$

[Ans.] ✓

Problem: (Example → 06)

$$\int x^2 \sqrt{x-1} \, dx$$

Soln:

$$\text{Let, } x-1 = u$$

$$dx = du$$

$$\text{and } x = u+1$$

$$\text{Then, } \int (u+1)^2 \sqrt{u} \, du$$

$$= \int (u^2 + 2u + 1) \sqrt{u} \, du$$

$$= \int u^{\frac{5}{2}} + 2u^{\frac{3}{2}} + u^{\frac{1}{2}} \, du$$

$$= \frac{2u^{\frac{7}{2}}}{\frac{7}{2}} + 2 \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{2}{7} (x-1)^{\frac{7}{2}} + \frac{4}{5} (x-1)^{\frac{5}{2}} + \frac{2}{3} (x-1)^{\frac{3}{2}} + C$$

[Ans.]

Example - 07) $\int_0^1 \tan^{-1} x \, dx$

$$= \tan^{-1} x \int dx - \int \left\{ \frac{d}{dx} (\tan^{-1} x) \int dx \right\} dx$$

$$= \tan^{-1} x \cdot x - \int \frac{1}{1+x^2} x \, dx$$

$$= x \tan^{-1} x - \int \frac{x \, dx}{1+x^2}$$

$$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x \, dx}{1+x^2}$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$$

$$\int \frac{f'(x)}{f(x)} = \ln(f(x))$$

Now, $\int_0^1 \tan^{-1} x \, dx$

$$= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$= \pi/4 - \ln \sqrt{2}$$

[Ans.]

Problem: Example → 08

$$\int \cos^4 x \, dx$$

Now, $\cos^4 x = (\cos^2 x)(\cos^2 x)$

$$[2\cos^2 \theta = 1 + \cos 2\theta]$$

Then, $\cos^4 x = \frac{1}{2}(1 + \cos 2x) \times \frac{1}{2}(1 + \cos 2x)$

$$\Rightarrow \cos^4 x = \frac{1}{4}(1 + 2\cos 2x + \cos^2 2x)$$

$$= \frac{1}{4} \left\{ (1 + 2\cos 2x) + \frac{1}{2}(1 + \cos 4x) \right\}$$

$$= \frac{1}{4} \left\{ (1 + 2\cos 2x) + \frac{1}{2} + \frac{1}{2}\cos 4x \right\}$$

$$= \frac{1}{4} + \frac{1}{2}\cos 2x + \frac{1}{8} + \frac{1}{8}\cos 4x$$

$$\therefore \cos^4 x = \frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$$

Now, $\int \cos^4 x \, dx = \int \left(\frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x \right) dx$

$$= \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$$

[Ans.]

Exercise → ch-7.2 (1-38)

Problem: 4) $\int x^2 e^{-2x} dx$

$$= x^2 \int e^{-2x} dx - \int \left\{ \frac{d}{dx} (x^2) \right\} \int e^{-2x} dx$$

$$= x^2 \frac{e^{-2x}}{-2} - \int 2x \cdot \frac{e^{-2x}}{-2} dx$$

$$= -\frac{1}{2} x^2 e^{-2x} + \int x e^{-2x} dx$$

$$= -\frac{1}{2} x^2 e^{-2x} + x \frac{e^{-2x}}{-2} - \int \left\{ \frac{d}{dx} (x) \right\} \int e^{-2x}$$

$$= -\frac{1}{2} x^2 e^{-2x} - \frac{x}{2} e^{-2x} - \int \frac{e^{-2x}}{-2} dx$$

$$= -\frac{1}{2} x^2 e^{-2x} - \frac{x}{2} e^{-2x} + \frac{1}{2} \frac{e^{-2x}}{(-2)} + C$$

$$= -\frac{1}{2} x^2 e^{-2x} - \frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} + C$$

[Ans.]

Problem: 7) $\int x^2 \cos x \, dx$

$$= x^2 \int \cos x \, dx - \int \left\{ \frac{d}{dx} (x^2) \int \cos x \, dx \right\} dx$$

$$= x^2 \sin x - \int 2x \sin x \, dx$$

$$= x^2 \sin x - 2 \left[x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (x) \int \sin x \, dx \right\} dx \right]$$

$$= x^2 \sin x - 2 \left[-x \cos x - \int -\cos x \, dx \right]$$

$$= x^2 \sin x + 2x \cos x - \int \cos x \, dx$$

$$= x^2 \sin x + 2x \cos x - \sin x + C$$

[Ans.]

[Prob.]

Problem: 11) $\int (\ln x)^2 \, dx$

$$= (\ln x)^2 \int dx - \int \left\{ \frac{d}{dx} (\ln x)^2 \int dx \right\} dx$$

$$= (\ln x)^2 x - \int 2 \ln x \cdot \frac{1}{x} x \, dx$$

$$= x (\ln x)^2 - \int 2 \ln x \, dx$$

$$= x (\ln x)^2 - 2 \frac{1}{x} + C$$

$$= x (\ln x)^2 - \frac{2}{x} + C \quad [\text{Ans.}]$$

Problem: 7) $\int x^2 \cos x \, dx$

$$= x^2 \int \cos x \, dx - \int \left\{ \frac{d}{dx} (x^2) \int \cos x \, dx \right\} dx$$

$$= x^2 \sin x - \int 2x \sin x \, dx$$

$$= x^2 \sin x - 2 \left[x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (x) \int \sin x \, dx \right\} dx \right]$$

$$= x^2 \sin x - 2 \left[-x \cos x - \int -\cos x \, dx \right]$$

$$= x^2 \sin x + 2x \cos x - \int \cos x \, dx$$

$$= x^2 \sin x + 2x \cos x - \sin x + C$$

[Ans.]

Problem: 11) $\int (\ln x)^2 \, dx$

$$= (\ln x)^2 \int dx - \int \left\{ \frac{d}{dx} (\ln x)^2 \int dx \right\} dx$$

$$= (\ln x)^2 x - \int 2 \ln x \cdot \frac{1}{x} x \, dx$$

$$= x (\ln x)^2 - \int 2 \ln x \, dx$$

$$= x (\ln x)^2 - 2 \frac{1}{x} + C$$

$$= x (\ln x)^2 - \frac{2}{x} + C$$

[Ans.]

Problem: 12) $\int \frac{\ln x}{\sqrt{x}} dx$

$$= \ln x \int \frac{1}{\sqrt{x}} dx - \int \left\{ \frac{d}{dx} (\ln x) \int \frac{1}{\sqrt{x}} dx \right\} dx$$

$$= \ln x \cdot \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} - \int \frac{1}{x} \cdot \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} dx$$

$$= 2 \ln x \cdot (1 - x^{-\frac{1}{2}}) - 2 \int \frac{1}{x} \cdot x^{-\frac{1}{2}} dx$$

$$= 2 \ln x (1 - x^{-\frac{1}{2}}) - 2 \int x^{-\frac{3}{2}} dx$$

$$= 2 \ln x (1 - x^{-\frac{1}{2}}) - 2 \left[\frac{x^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} \right] + C$$

$$= 2 \ln x (1 - x^{-\frac{1}{2}}) + 4 x^{-\frac{1}{2}} + C$$

[Ans.]

Problem: 14) $\int \ln(x^2 + 4) dx$

$$= \ln(x^2 + 4) \int dx - \int \left\{ \frac{d}{dx} (\ln(x^2 + 4)) \int dx \right\} dx$$

$$= x \ln(x^2 + 4) - \int \frac{1}{x^2 + 4} (2x) x dx$$

$$= x \ln(x^2 + 4) - \int 2x^2 (x^2 + 4)^{-1} dx$$

$$= x \ln(x^2 + 4) - 2 \int x^2 (x^2 + 4)^{-1} dx$$

$$\begin{aligned}
 \text{Now, } & \int \frac{x^2}{x^2+4} dx \\
 &= \int \left(1 - \frac{4}{x^2+4} \right) dx \\
 &= \int dx - 4 \int \frac{dx}{x^2+4} \\
 &= x - 4 \cdot \frac{1}{2} \tan^{-1} \frac{x}{2} + C \\
 &= x - 2 \tan^{-1} \left(\frac{x}{2} \right) + C
 \end{aligned}$$

$\left[\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C \right]$

[Ans.]

$$\begin{aligned}
 \text{Then, } & \int \ln(x^2+4) dx \\
 &= x \ln(x^2+4) - 2x + 4 \tan^{-1} \left(\frac{x}{2} \right) + C
 \end{aligned}$$

[Ans.]

Problem: 16) $\int \cos^{-1}(2x) dx$

$$\begin{aligned}
 &= \cos^{-1}(2x) \int dx - \int \left\{ \frac{d}{dx} (\cos^{-1}(2x)) \int dx \right\} dx \\
 &= x \cos^{-1}(2x) - \int -\frac{2}{\sqrt{1-4x^2}} \cdot x dx \\
 &= x \cos^{-1}(2x) + 2 \int \frac{x}{\sqrt{1-4x^2}} dx \\
 &= x \cos^{-1}(2x) + 2 \left[\frac{(\sqrt{1-4x^2})^{-1}}{-1/2} \int x dx - \int \frac{d}{dx} (\sqrt{1-4x^2})^{-1} \int x dx \right] dx
 \end{aligned}$$

$$= x \cos^{-1} 2x + x^2 (\sqrt{1-4x^2})^{-1} + \int \sqrt{1-4x^2}$$

$$\int \frac{2x}{\sqrt{1-4x^2}} dx$$

$$\text{let, } 1-4x^2 = u$$

$$\Rightarrow -8x dx = du$$

$$\therefore x dx = \frac{du}{-8}$$

$$\text{Then, } 2 \int \frac{\frac{du}{-8}}{\sqrt{u}}$$

$$\Rightarrow \frac{-1}{4} \int u^{-1/2} du$$

$$= \frac{-1}{4} \frac{u^{-1/2+1}}{1-1/2} + C$$

$$= -1/2 (1-4x^2)^{-1/2} + C$$

$$\text{Then, } \int \cos^{-1}(2x) dx$$

$$= x \cos^{-1}(2x) - \frac{1}{2} (1-4x^2)^{1/2} + C$$

[Ans.]

$$\frac{1}{1+x^2} - 1 \cdot \frac{1}{1+x^2} = \frac{1-x^2}{1+x^2} \quad \text{LIATE}$$

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Problem: 18) $\int x \tan^{-1} x \, dx$

$$= \tan^{-1} x \int x \, dx - \int \left\{ \frac{d}{dx} (\tan^{-1} x) \right\} \int x \, dx \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[\int \frac{1}{1+x^2} \, dx - \int \frac{1}{1+x^2} \, dx \right]$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[\int dx - \int \frac{dx}{1+x^2} \right]$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C$$

[Ans.]

Problem: 21) $\int \sin(\ln x) \, dx$

$$= \int \sin(\ln x) \, dx - \int \left\{ \frac{d}{dx} (1) \right\} \int \sin(\ln x) \, dx$$

$$= \int \sin(\ln x) \, dx - 0$$

$$= \int \sin(\ln x) \, dx$$

$$= -\cos(\ln x)$$

$$= -x \cos(\ln x)$$

Problem: 21) $\int \sin(\ln x) dx$

Let, $\ln x = u$

$$\Rightarrow e^{\ln x} = e^u$$

$$\therefore x = e^u$$

$$dx = e^u du$$

$$= \int \sin(u) x^u du$$

$$= \sin u \int e^u du - \int \left(\frac{d}{du} (\sin u) \right) \int e^u du du$$

$$= \sin u e^u - \int e^u (\cos u) du$$

$$= e^u \sin u - \int e^u \cos u du$$

$$= e^u \sin u - \left[\cos u e^u - \int \left(\frac{d}{du} (\cos u) \right) \int e^u du du \right]$$

$$= e^u \sin u - \left[e^u \cos u - \int (-\sin u) e^u du \right]$$

$$= e^u \sin u - e^u \cos u + \int e^u \sin u du$$

Let, $\int \sin(u) e^u du = I$

So, $2I = e^u \sin u - e^u \cos u + c$

$$\therefore I = \frac{1}{2} [e^u \sin u - e^u \cos u] + c$$

$$= \frac{1}{2} [e^{\ln x} \sin(\ln x) - e^{\ln x} \cos(\ln x)]$$

$$= \frac{1}{2} x [\sin(\ln x) - \cos(\ln x)] + c \quad [\text{Ans.}]$$

Problem: 24) $\int x \tan^2 x \, dx$

$$= \int x (\sec^2 x - 1) \, dx$$

$$= x \int (\sec^2 x - 1) \, dx - \int \frac{d}{dx} (x) \int (\sec^2 x - 1) \, dx$$

$$= x \tan x - x - \int (\sec^2 x - 1) \, dx (\tan x - x) \, dx$$

$$\checkmark = x \tan x - x - \ln |\sec x| - \frac{x^2}{2} + C$$

$$\checkmark = x \tan x - \frac{1}{2} x^2 + \ln |\cos x| + C$$

$$\begin{aligned} \left[\because \int \tan x \, dx &= -\ln |\cos x| + C \right. \\ &= \ln |\sec x| + C \end{aligned}$$

[Ans.]

Problem: 26) $\int \frac{x e^x \, dx}{(x+1)^2}$

$$= \int x e^x (x+1)^{-2} \, dx$$

$$= \frac{(x+1)^{-2}}{\frac{d}{dx}} \int x e^x \, dx - \int \frac{d}{dx} (x+1)^{-2} \int x e^x \, dx \, dx$$

$$= \frac{(x+1)^{-2}}{\frac{d}{dx}}$$

$$x+1 = u$$

→

Problem:

26)

$$\int \frac{x e^x}{(1+x)^2} dx$$

$$\text{Let, } I = \int \frac{x e^x dx}{(1+x)^2}$$

$$= \int \frac{(x+1-1) e^x}{(x+1)^2} dx$$

$$= \int \left\{ \frac{e^x}{(x+1)^2} - \frac{e^x}{(x+1)^2} \right\} dx$$

$$\text{Let, } x+1 = t \quad \therefore dx = dt$$

$$x = t-1$$

$$\text{Now, } \int \left\{ \frac{e^{t-1}}{t} - \frac{e^{t-1}}{t^2} \right\} dt$$

$$= \bar{t}^1 \int e^{t-1} dt - \int \left\{ \frac{d}{dt} (\bar{t}^1) \int e^{t-1} dt \right\}$$

$$- \left[\bar{t}^2 \int e^{t-1} dt - \int \left\{ \frac{d}{dt} (\bar{t}^2) \int e^{t-1} dt \right\} d\bar{t} \right]$$

$$= \bar{t}^1 e^{t-1} - \int -\bar{t}^{-2} \cdot e^{t-1} - \left[\bar{t}^2 e^{t-1} - \int -2\bar{t}^3 e^{t-1} dt \right]$$

$$= \bar{t}^1 e^{t-1} + \int \bar{t}^2 e^{t-1} - \bar{t}^2 e^{t-1} - 2 \int \bar{t}^3 e^{t-1} dt$$

$$\begin{aligned}
 & \int t^2 e^{t-1} \\
 &= t^2 \int e^{t-1} dt - \int \{2t^{-3} \int e^{t-1} dt\} dt \\
 &= t^2 e^{t-1} + 2 \int t^3 e^{t-1} dt
 \end{aligned}$$

Now,

$$\begin{aligned}
 & t^1 e^{t-1} + \cancel{t^2 e^{t-1}} - \cancel{t^2 e^{t-1}} - 2 \int \cancel{t^3 e^{t-1}} dt \\
 & \quad + 2 \int \cancel{t^3 e^{t-1}} dt + C
 \end{aligned}$$

$$= t^1 e^{t-1} + C$$

$$= \frac{e^{t-1}}{t} + C$$

$$= \frac{e^x}{1+x} + C$$

[Ans.]

Problem: 30) $\int_{\sqrt{e}}^e \frac{\ln x}{x^2} dx$

Now, $\int \ln x \cdot x^{-2} dx$

$$= \ln x \int x^{-2} dx - \int \left\{ \frac{d}{dx} (\ln x) \int x^{-2} dx \right\} dx$$

$$= \ln x \frac{x^{-2+1}}{-2+1} - \int \frac{1}{x} \frac{x^{-2+1}}{-2+1} dx$$

$$= -\ln x \cdot x^{-1} + \int \frac{x^{-1}}{x} dx$$

$$= -\ln x \cdot \frac{1}{x} + \int \frac{1}{x^2} dx$$

$$= -\frac{\ln x}{x} + \frac{x^{-2+1}}{-2+1}$$

$$= -\frac{1}{x} \ln x - \frac{1}{x} + C$$

Now, $\left[-\frac{1}{x} \ln x - \frac{1}{x} \right]_{\sqrt{e}}^e$

$$= -\frac{1}{e} \ln e - \frac{1}{e} - \left[-\frac{1}{\sqrt{e}} \ln(\sqrt{e}) - \frac{1}{\sqrt{e}} \right]$$

$$= -\frac{1}{e} - \frac{1}{e} + \frac{1}{\sqrt{e}} \cdot \frac{1}{2} + \frac{1}{\sqrt{e}}$$

$$= \frac{3\sqrt{e} - 4}{2e} \quad [\text{Ans.}]$$

$+\sqrt{e} \cdot \frac{1}{2}$

$$x^{\frac{1}{2}} \cdot x = x^{\frac{3}{2}}$$

$$\frac{3}{2} - \frac{1}{2} = 1$$

$$\frac{3-1}{2} = 1$$

$$\frac{2}{2} = 1$$

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Problem: 37) $\int_1^3 \sqrt{x} \tan^{-1} \sqrt{x} dx$

$$\int \sqrt{x} \tan^{-1}(\sqrt{x}) dx$$

$$= \tan^{-1} \sqrt{x} \int \sqrt{x} dx - \int \left\{ \frac{d}{dx} (\tan^{-1} \sqrt{x}) \int \sqrt{x} dx \right\} dx$$

$$= \tan^{-1} \sqrt{x} \cdot \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} - \int \frac{1}{1+x} \cdot \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} \cdot \frac{1}{2\sqrt{x}} dx$$

$$= \frac{2 \tan^{-1} \sqrt{x}}{3} x^{\frac{3}{2}} - \frac{2}{3} \int \frac{x^{\frac{3}{2}}}{1+x} dx$$

$$= \frac{2}{3} \tan^{-1} \sqrt{x} x^{\frac{3}{2}} - \frac{1}{3} \int \left(1 - \frac{1}{1+x} \right) dx$$

$$= \left[\frac{2}{3} \tan^{-1} \sqrt{x} x^{\frac{3}{2}} - \frac{1}{3} x + \frac{1}{3} \ln |1+x| + C \right]_1^3$$

$$= \frac{2}{3} \tan^{-1} \sqrt{3} (3)^{\frac{3}{2}} - 1 + \frac{1}{3} \ln 4 \quad [\text{Ans}]$$

$$= \frac{2}{3} \tan^{-1} \sqrt{3} (3)^{\frac{3}{2}} - 1 + \frac{1}{3} \ln 4 - \left(\frac{2}{3} \tan^{-1} (1) + \frac{1}{3} - \frac{1}{3} \ln 2 \right)$$

$$= \frac{2}{3} \cdot \frac{\pi}{3} (3)^{\frac{3}{2}} - 1 + \frac{1}{3} \ln 4 - \frac{2}{3} \cdot \frac{\pi}{6} + \frac{1}{3} - \frac{1}{3} \ln 2$$

$$= \frac{2}{3} \cdot \frac{\pi}{3} (3)^{\frac{3}{2}} - \frac{2}{3} + \frac{1}{3} \ln 4 - \frac{2}{3} \cdot \frac{\pi}{6} - \frac{1}{3} \ln 2$$

$$= \frac{12}{3} \left[\frac{2\pi}{3} 3^{\frac{3}{2}} - \frac{2}{3} + \frac{1}{3} \ln 4 - \frac{\pi}{6} - \frac{1}{3} \ln 2 \right]$$

$$= \frac{2}{3} \left[\pi \sqrt{3} - 1 + 2 \ln 4 - \frac{\pi}{4} \right]$$

$$= 2\pi\sqrt{3} - 2 - \frac{\pi}{2} = \frac{1}{3} [2\pi\sqrt{3} - 2 + \ln 2 - \frac{\pi}{2}]$$

Problem: 32) $\int_0^{\frac{\sqrt{3}}{2}} \sin^{-1} x \, dx$

$$= \sin^{-1} x \int dx - \int \left\{ \frac{d}{dx} (\sin^{-1} x) \right\} x \, dx$$

$$= x \sin^{-1} x - \int \frac{1}{\sqrt{1-x^2}} x \, dx$$

$$= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} \, dx$$

$$= x \sin^{-1} x - \int \frac{\frac{du}{-2}}{\sqrt{u}} \quad \cancel{\sqrt{1-x^2}}$$

$$\begin{cases} 1-x^2 = u \\ -2x \, dx = du \\ \therefore x \, dx = \frac{du}{-2} \end{cases}$$

$$= x \sin^{-1} x + \frac{1}{2} \int u^{-1/2} \, du$$

$$= x \sin^{-1} x + \frac{1}{2} \frac{u^{-1/2+1}}{-1/2+1} + C$$

$$= \left[x \sin^{-1} x + (1-x^2)^{1/2} + C \right]_0^{\sqrt{3}/2}$$

$$= \frac{\sqrt{3}}{2} \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) + \left(1 - \frac{3}{4} \right)^{1/2} - 1 + C$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{\pi}{3} + \left(\frac{1}{4} \right)^{1/2} - 1 + C$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{\pi}{3} + \frac{1}{2} - 1 + C$$

$$= \frac{\pi \sqrt{3}}{6} + \frac{1}{2} + C$$

[Ans.]

Problem: 38) $\int_0^2 \ln(x^2+1) dx$

$$= \ln(x^2+1) \int dx - \int \left\{ \frac{d}{dx} (\ln(x^2+1)) \right\} \int dx \} dx$$

$$= x \ln(x^2+1) - \int \frac{2x}{x^2+1} dx$$

$$= x \ln(x^2+1) - 2 \int \frac{x}{x^2+1} dx$$

$$= x \ln(x^2+1) - 2 \left[\int \left(1 - \frac{1}{x^2+1} \right) dx \right]$$

$$= \left[x \ln(x^2+1) - 2x + 2 \tan^{-1}(x) \right]_0^2$$

$$= 2 \ln(5) - 4 + 2 \tan^{-1}(2) + C$$

[Ans.]

Date: 21.08.2023

(class → 21)

(sample question)

Question: 1)

a) from January 5, 2004 to August 30, 2005

Average Rate of change, V_{ave}

$$= \frac{68.81 - 33.70}{608 - 5}$$

$$= 0.05822$$

b) from August 30, 2005 to April 20, 2006

$$V_{ave} = \frac{71.95 - 68.81}{841 - 608}$$

$$= 0.013476$$

Question: 2)

$$s(t) = t^2 - 2$$

$$V_{ave} = \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

$$= \frac{f(3) - f(2)}{3 - 2}$$

$$= \frac{(3^2 - 2) - (2^2 - 2)}{1}$$

$$= 9 - 2 - 4 + 2$$

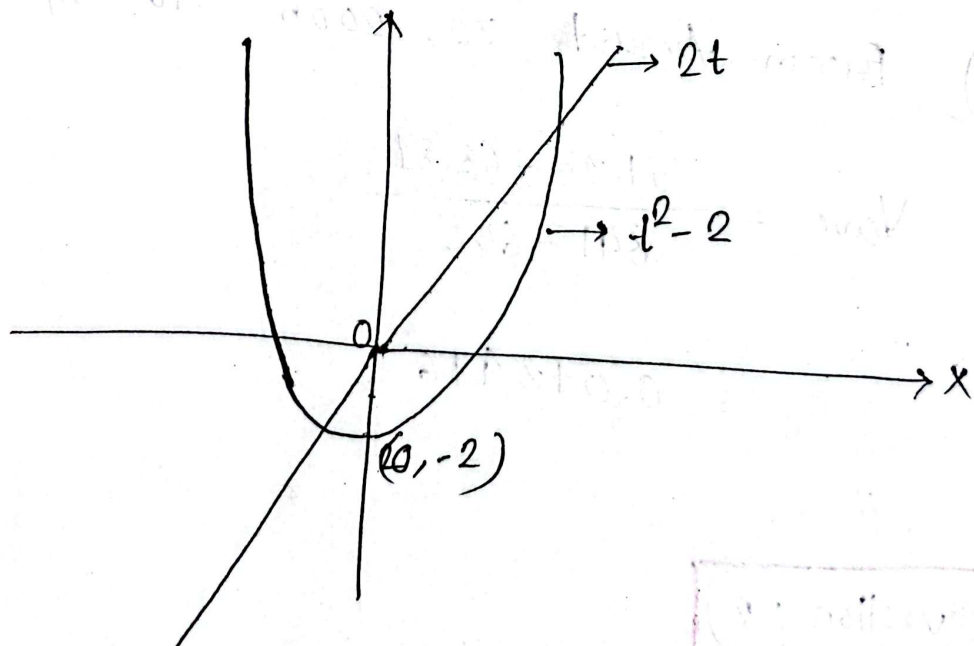
$$= 5 \text{ m s}^{-1}$$

b) $V_{ins} = \frac{d}{dt} (t^2 - 2)$
 $= 2t$

At time, $t = 2$, $V_{ins} = 2 \times 2 = 4 \text{ m/s}$

c) position function $s(t) = t^2 - 2$

velocity function $v(t) = 2t$



Question: 3)

a) $f(x) = x^3 + 3$

b) $g(x) = 6x - x^2$

(i) ins rate of change, $f'(x) = \frac{d}{dx} (x^3 + 3)$
 $= 3x^2 + 0$
 $= 3x^2$

$g'(x) = 6 - 2x$

OR,

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3 - 3 + 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3 - 3 + 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2) - 3 + 3}{h}$$

$$= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2$$

$$= 3x^2$$

$$(ii) \quad f'(x) \Big|_{x=1} = 3x^2 = 3(1)^2 = 3$$

$$\text{and} \quad g'(x) \Big|_{x=1} = 6 - 2 \cdot (1) = 6 - 2 = 4$$

(iii) tangent line equation:

$$y - y_1 = m(x - x_1)$$

$$\text{for } f(x), \quad m = 3$$

$$\text{At, } x = 1, \quad y = x^3 + 3 = (1)^3 + 3 = 4$$

$$\text{Then, } y - 4 = 3(x - 1)$$

$$\Rightarrow \boxed{y - 3x = 1}$$

$$\text{for, } g(x), \quad m = 4$$

$$\text{At, } x = 1, \quad y = 6x - x^2 = 6 \cdot 1 - (1)^2 = 6 - 1 = 5$$

$$\text{Then, } y - 5 = 4(x - 1)$$

$$\therefore \boxed{y - 4x = 1}$$

(iv) For $f(x)$, slope of secant line, $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$

$$= \frac{f(-1) - f(1)}{1 - (-1)}$$

$$= \frac{(-1)^3 + 3 - 1^3 - 3}{2}$$

$$= \frac{-1 - 1}{2}$$

$$m_{sec} = -1$$

For $g(x)$, $m_{sec} = \frac{f(-1) - f(1)}{1 - (-1)}$

$$= \frac{6(-1) - (-1)^2 - 6 + 1}{2}$$

$$= \frac{-6 - 6}{2}$$

$$= \frac{-12}{2}$$

$$m_{sec} = -6$$

Equation of secant line, $y - y_1 = m(x - x_1)$

For $f(x)$,
At $x = 1$, $y = 1^3 + 3 = 4$

$$y - 4 = -1(x - 1)$$

$$\therefore y + x = 5$$

For $g(x)$
At $x = 1$, $y = 6 \cdot 1 - 1^2 = 5$

$$y - 5 = -6(x - 1)$$

$$\therefore y + 6x = 11$$

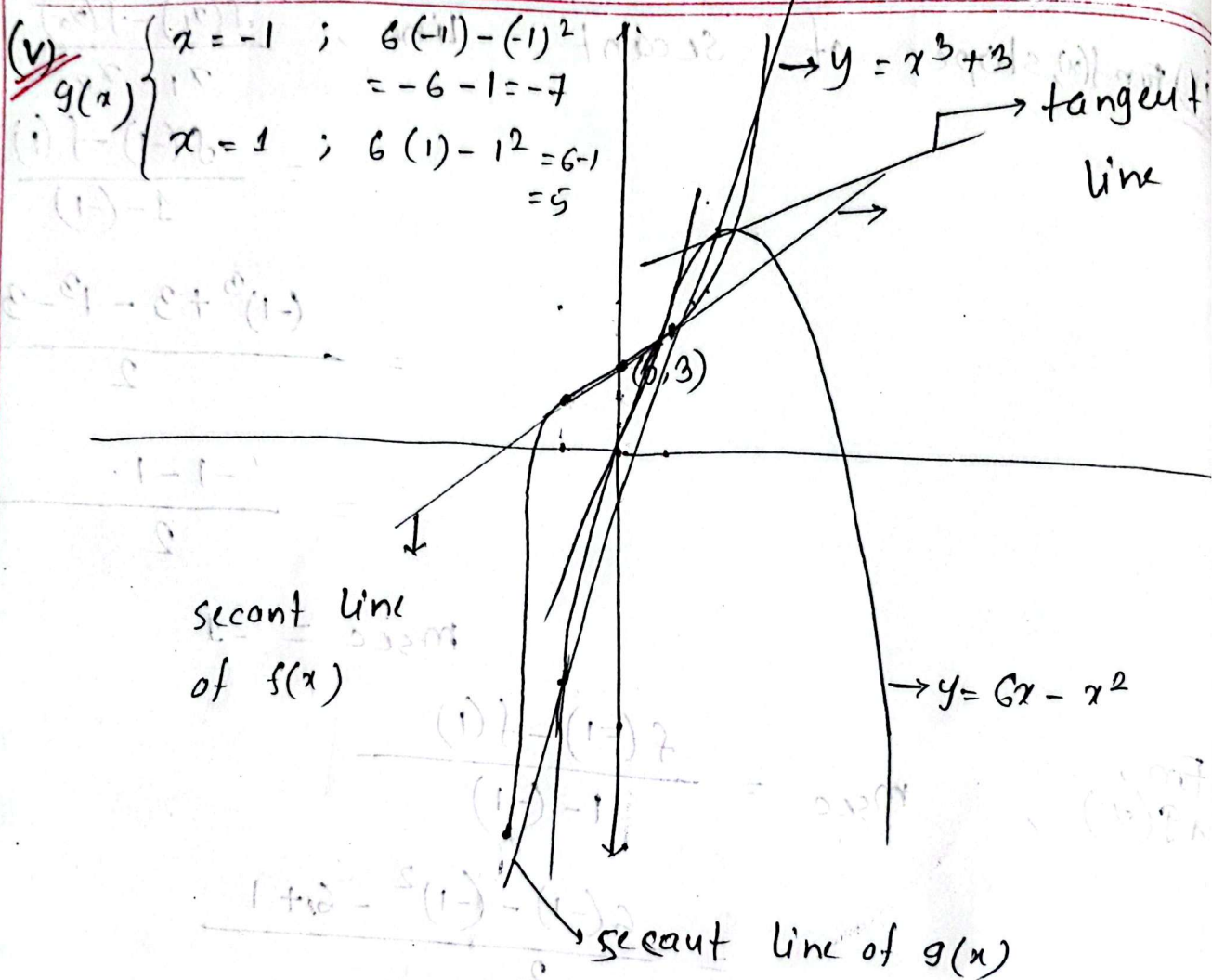
[Ans.]

$$f(x) \begin{cases} x = -1 ; y = (-1)^3 + 3 = -1 + 3 = 2 \\ x = 1 ; y = 1^3 + 3 = 4 \end{cases}$$

$$y = x^3 + 3$$

$$x = 0 ; y =$$

$$(v) \begin{cases} x = -1 ; 6(-1) - (-1)^2 = -6 - 1 = -7 \\ x = 1 ; 6(1) - 1^2 = 6 - 1 = 5 \end{cases}$$



$$f(x) = 4 \quad \text{At } x = 1$$

$$g(x) = 5 \quad \text{At } x = 1$$

Question: 4)

a) $f(x) = -3x^2$

b) $g(x) = -4 + x^3$

(i) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{-3(x+h)^2 + 3x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{-3(x^2 + 2xh + h^2) + 3x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{-6xh - 3h^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{h(-6x - 3h)}{h}$$
$$= -6x$$

and $g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{-4 + (x+h)^3 + 4 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-4 + x^3 + 3x^2h + 3xh^2 + h^3 + 4 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h}$$

$$= 3x^2$$

$$(ii) \quad f'(x) \Big|_{x=-1} = -6(-1)^2 = -6 = m_{tan}$$

$$\text{At } x = -1, \quad y = -3(1)^2 = -3$$

$$y - (-3) = -6(x + 1)$$

$$\Rightarrow y + 6x = -6 - 3 = -9$$

$$\therefore y + 6x = -9$$

$$\text{and } g'(x) \Big|_{x=-1} = 3(-1)^2 = 3 = m_{tan}$$

$$\text{At } x = -1, \quad y = -4 + (-1)^3 = -4 - 1 = -5$$

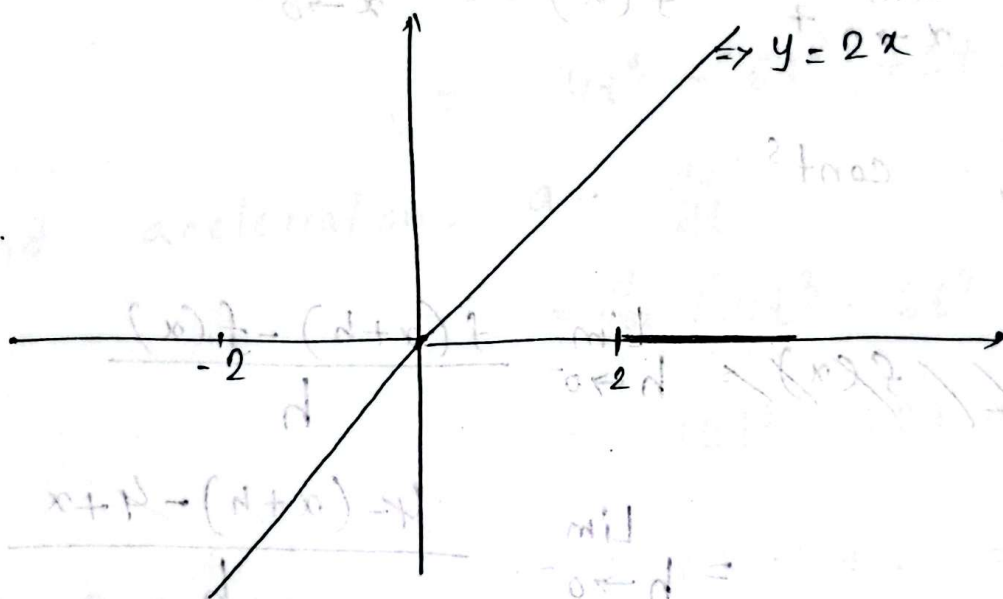
$$y - (-5) = 3(x + 1)$$

$$\therefore y = 3x - 2$$

$$\text{[Ans.]}$$

(5) Equation : $y = x^2$ and $y = 4$

$$f(x) = y = \begin{cases} x^2 & ; -2 \leq x \leq 2 \\ 4 & ; \cancel{x < -2} \quad 2 \leq x \leq 4 \end{cases}$$



(a) $g(x)$ increasing $\Rightarrow 0 \leq x \leq 2$
 decreasing $\Rightarrow -2 \leq x \leq 0$
 constant $\Rightarrow 2 \leq x \leq 4$

(b) $g(x)$ never negative

Question: 6)

a) $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 4 - x = 4$

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 4 - x^2 = 4$

so, cont^s.

$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0^-} \frac{4 - (x+h) - 4 + x}{h}$

$= \lim_{h \rightarrow 0^-} \frac{-x - h + x}{h}$

$= 0$

and $\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0^+} \frac{4 - (x+h)^2 - (4 - x^2)}{h}$

$= \lim_{h \rightarrow 0^+} \frac{4 - (x^2 + 2xh + h^2) - 4 + x^2}{h}$

$= \lim_{h \rightarrow 0^+} \frac{-2xh - h^2}{h} = \lim_{h \rightarrow 0^+} \frac{-2x - h}{1} = -2x$

so, differentiable.

Question: 7)

$$s = t^4 - 2t^3 + t^2 - t$$

a) velocity, $v = \frac{ds}{dt}$

$$\Rightarrow v = \frac{d}{dt} (t^4 - 2t^3 + t^2 - t)$$
$$= 4t^3 - 6t^2 + 2t - 1$$

and acceleration, $a = \frac{dv}{dt}$

$$= \frac{d}{dt} (4t^3 - 6t^2 + 2t - 1)$$
$$= 12t^2 - 12t + 2$$

b) $a = \frac{dv}{dt} \Big|_{t=1} = 12 - 12 + 2 = 2 \text{ m s}^{-2}$

[Ans]

Question: 08) a) Given that,

$$f(x) = 2 \sin x + \sin^2 x$$

tangent line horizontal,

$$f'(x) = 0$$

$$\Rightarrow 2 \cos x + 2 \sin x \cos x = 0$$

$$\Rightarrow \cos x (2 + 2 \sin x) = 0$$

$$\Rightarrow \cos x (1 + \sin x) = 0$$

$$\cos x = 0$$

$$\text{and } \sin x = -1$$

$$x = \frac{\pi}{2} + 2n\pi$$

$$\text{and } \frac{3\pi}{2} + 2n\pi$$

$$(4n-1) \frac{\pi}{2}$$

(6g) (b) If the slope of one line is the negative Reciprocal of the 2nd line, Lines are perpendicular.

$$6x + 2y = 1$$

$$\Rightarrow 2y = 1 - 6x$$

$$\Rightarrow y = \frac{-6x + 1}{2}$$

$$y = -3x + \frac{1}{2}$$

$$m = -3$$

$$y = mx + c$$

↓
slope

We need to find where the curve $f(x) = \sqrt{1+2x}$ has a tangent line with slope of $1/3$. That's opposite of -3 .

$$f'(x) = \frac{1}{2} (1+2x)^{-1/2} \cdot 2 = \frac{1}{\sqrt{1+2x}}$$

Then, $\frac{1}{\sqrt{1+2x}} = \frac{1}{3}$

$$3 = \sqrt{1+2x}$$

$$\Rightarrow 9 = 1+2x$$

$$\Rightarrow 8 = 2x$$

$$x = 4$$

$$y = \sqrt{1+(2 \times 4)}$$

$$= \sqrt{1+8}$$

$$= \sqrt{9}$$

$$= 3$$

$$\therefore (x, y) = (4, 3)$$

Problem: 9)

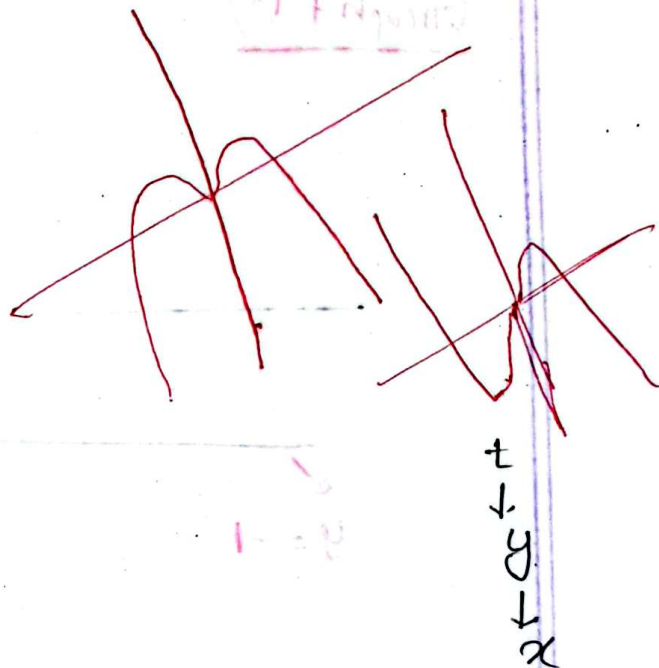
(a) $\overset{\text{sin} x}{(a)} = \text{I} \rightarrow \text{cos} x$

(b) $\overset{\text{+ve slope, -ve slope}}{(b)} = \text{IV (constant)}$

(c) = (I)

(d) = (III)

$y = ax$
 $\downarrow y = a$



Question: 10)

$$t = \ln(\sin(y^2 + 1))$$

$$y = \cos(x^2 e^x)$$

Then, $\frac{dt}{dx} = \frac{d}{dx} (\ln(\sin(y^2 + 1)))$

$$= \frac{dt}{dy} \cdot \frac{dy}{dx}$$

$$= \frac{d}{dy} \cos(x^2 e^x) \cdot \frac{d}{dx} \ln(\sin(y^2 + 1))$$

$$= -\sin(x^2 e^x) (2x e^x + e^x x^2) \cdot \frac{1}{\sin(y^2 + 1)}$$

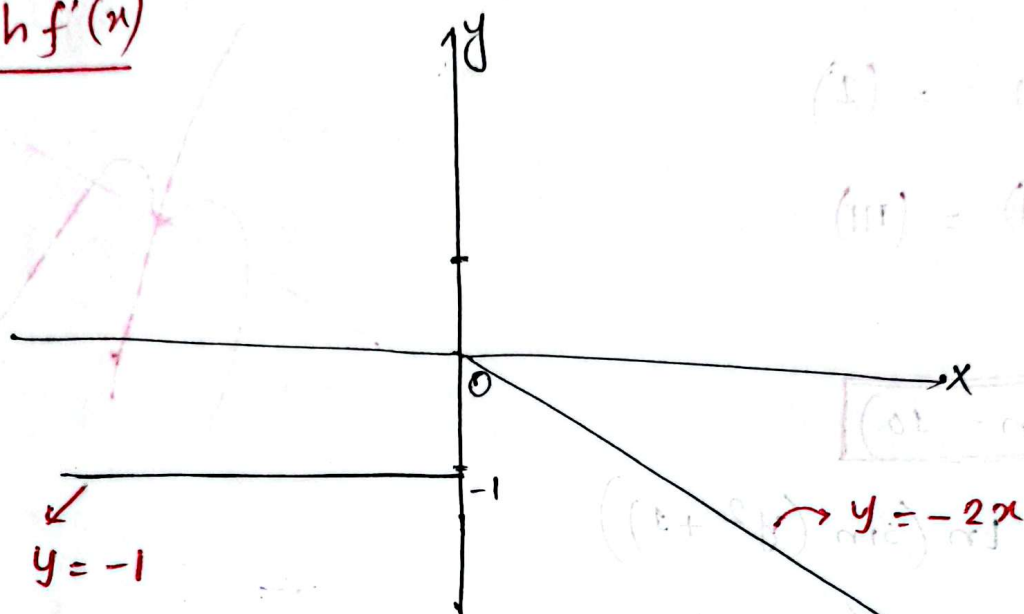
$$= -2y \sin(x^2 e^x) x e^x (2 + x) \cdot \frac{\cos(y^2 + 1) \cdot 2y}{\cot(y^2 + 1)}$$

[Ans.]

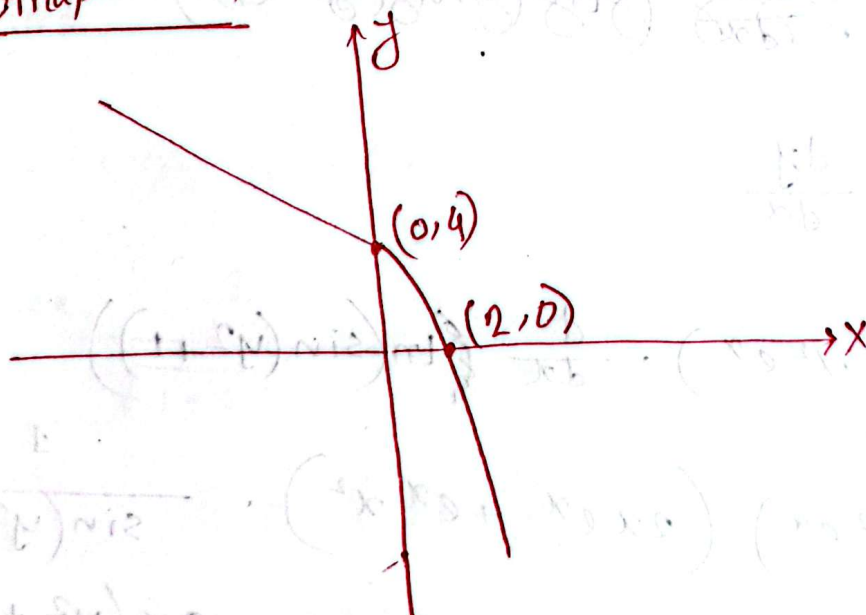
(6) (c)

$$f'(x) = \begin{cases} -1 & ; x < 0 \\ -2x & ; x > 0 \end{cases}$$

Graph $f'(x)$



Graph $f(x)$





Sample Question (Final)

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(Class: 22)

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Problem: 11) a) inner function = $\cos x$
and outer function = $\ln(\cos x)$

$$f(g(x)) = \ln(\cos x)$$

Now,

$$\frac{dy}{dx} = \frac{d}{dx} (\ln(\cos x))$$
$$= \frac{1}{\cos x} (-\sin x)$$

$$\therefore \frac{dy}{dx} = -\tan x \quad [\text{Ans.}]$$

(b) inner function = $e^{2x} + 1$

Outer function = $\sqrt[3]{1+e^{2x}}$

$$f(g(x)) = \sqrt[3]{1+e^{2x}}$$

$$\frac{dy}{dx} = \frac{d}{dx} (\sqrt[3]{1+e^{2x}})$$
$$= \frac{1}{3} (1+e^{2x})^{\frac{1}{3}-1} (2e^{2x}) = \frac{2}{3} (1+e^{2x})^{-\frac{2}{3}} e^{2x}$$

[Ans.]

(c) inner function = $x^5 + 3x^2 + 1$
 outer function = $(x^5 + 3x^2 + 1)^{-10}$

Then, $\frac{dy}{dx} = -10 (x^5 + 3x^2 + 1)^{-11} (5x^4 + 6x)$

[Ans.]

✓ Problem: 12)

Given that,

$$y = \cos(\cos x)$$

Now, $y' = \frac{dy}{dx}$
 $= \frac{d}{dx} (\cos(\cos x))$
 $= -\sin(\cos x) \cdot (-\sin x)$

$$\therefore y' = \sin(\cos x) \sin x$$

Now, $m_{\tan} = \sin(\cos x) \sin x$

Then, $y - y_1 = m(x - x_1)$

At, $x = \pi/2$, $m_{\tan} = 0$

and $y = \cos(\cos \pi/2) = \cos(0) = 1$

so, eqn of tangent line: $y - 1 = 0$

[Ans.]

blum: 13)

(a) $f(x) = \sec^3\left(\frac{x}{\sqrt{1-x^2}}\right)$

Now, $f'(x) = 3 \sec^2\left(\frac{x}{\sqrt{1-x^2}}\right) \cdot \frac{\frac{d}{dx}(x) - x \frac{d}{dx}(\sqrt{1-x^2})}{1-x^2}$

$$\frac{\sqrt{1-x^2} - x \frac{d}{dx}(\sqrt{1-x^2})}{1-x^2}$$

$$= 3 \sec^2\left(\frac{x}{\sqrt{1-x^2}}\right) \cdot \frac{\sqrt{1-x^2} - x \frac{1}{2\sqrt{1-x^2}}(-2x)}{1-x^2}$$

$$= 3 \sec^2\left(\frac{x}{\sqrt{1-x^2}}\right) \cdot \frac{\sqrt{1-x^2} + \frac{x^2}{\sqrt{1-x^2}}}{1-x^2}$$

$$= 3 \sec^2\left(\frac{x}{\sqrt{1-x^2}}\right) \cdot \frac{\frac{1-x^2 + x^2}{\sqrt{1-x^2}}}{1-x^2}$$

$$= 3 \sec^2\left(\frac{x}{\sqrt{1-x^2}}\right) \cdot \frac{1}{(1-x^2)\sqrt{1-x^2}}$$

[Ans.]

(b)

$$g(x) = (x^7 - 3)^{-2} \cdot \tan\left(\frac{3}{x}\right)$$

$$= (x^7 - 3)^{-2} \sec^2\left(\frac{3}{x}\right) \cdot \frac{-1}{x^2} + \tan\left(\frac{3}{x}\right) \cdot (-2)(x^7 - 3)^{-3} \cdot 7x^2$$

$$= -\frac{3}{x^2} (x^7 - 3)^{-2} \sec^2\left(\frac{3}{x}\right) - 14x^2 (x^7 - 3)^{-3} \tan\left(\frac{3}{x}\right)$$

[Ans.]

✓ Problem: 14) $F(x) = f(g(x))$

$$\text{Now, } f'(x) = f'(g(x)) \cdot g'(x)$$

$$\Rightarrow F'(2) = f'(g(2)) \cdot g'(2)$$

$$= f'(1) \cdot (-5)$$

$$= -5 f'(1)$$

$$G(x) = g(f(x))$$

$$G'(x) = g'(f(x)) \cdot f'(x)$$

$$\Rightarrow G'(2) = g'(f(2)) \cdot f'(2)$$

$$= g'(0) \cdot 4$$

$$= 4 g'(0)$$



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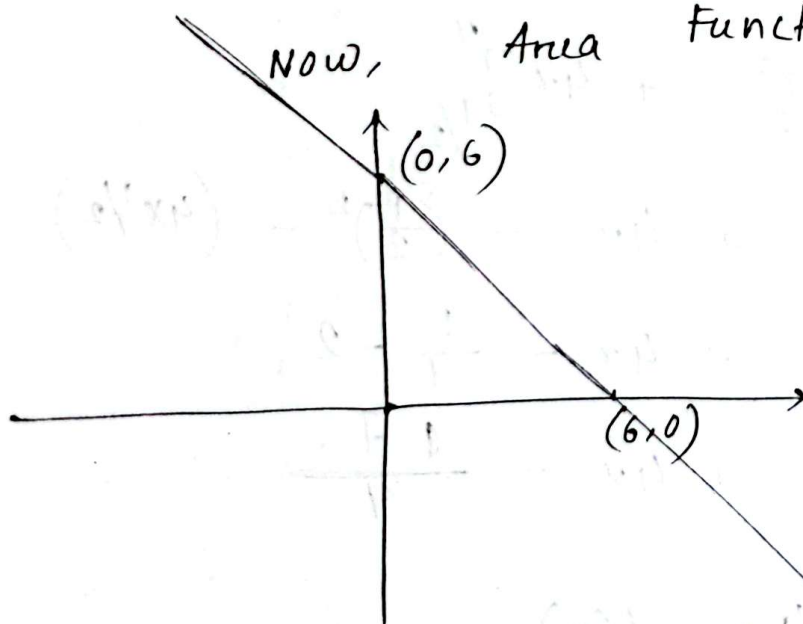
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Problem: 15) (a) $f(x) = 6 - x$; $[a, x] = [-2, x]$



$$A(x) = \int_{-2}^x (6 - x) dx$$

$$= \left[6x - \frac{x^2}{2} \right]_{-2}^x$$

$$= 6x - \frac{x^2}{2} - 6(-2) + \frac{(-2)^2}{2}$$

$$= 6x - \frac{x^2}{2} + 12 + 2$$

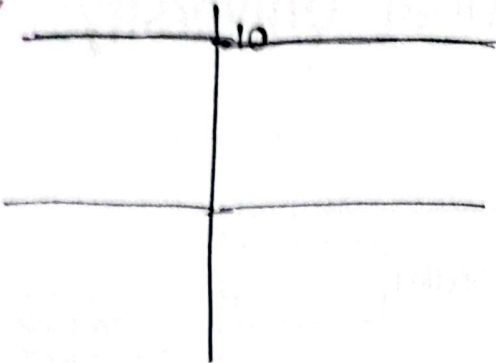
$$A(x) = 6x - \frac{x^2}{2} + 24$$

$$A'(x) = 6 - \frac{1}{2} \cdot 2x$$

$$= 6 - x$$

$$A'(x) = f(x) \quad [\text{confirmed}]$$

(b) Area function, $A(x) = \int_2^x 10$



$$= [10x]_2^x$$

$$= 10[x-2]$$

$$A'(x) = 10 = f(x)$$

(c) $A(x) = \int_{1/2}^x (2x+4) dx$

$$= \left[2 \frac{x^2}{2} + 4x \right]_{1/2}^x$$

$$= x^2 + 4x - \left(\frac{1}{2}\right)^2 - (4 \cdot \frac{1}{2})$$

$$= x^2 + 4x - \frac{1}{4} - 2$$

$$= x^2 + 4x - \frac{9}{4}$$

$$A'(x) = 2x + 4 = f(x)$$

Problem 16)

$$y = x^3 - 1$$

$$\int_0^2 (x^3 - 1) dx$$

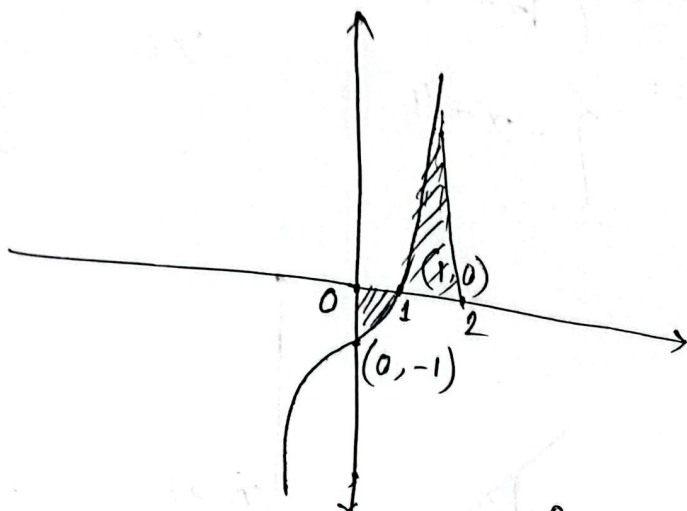
$$= \left[\frac{x^4}{4} - x \right]_0^2$$

$$= \frac{1}{4} [2^4 - 2]$$

$$= 4 - \frac{1}{2}$$

$$= \frac{8-1}{2}$$

$$= \frac{7}{2} \text{ [Ans.]}$$



$$\int_0^1 f(x) dx + \int_1^2 f(x) dx$$

$$= \int_0^1 -x^3 dx + \int_1^2 x^3 dx$$

$$= \left[-\frac{x^4}{4} \right]_0^1 + \left[\frac{x^4}{4} \right]_1^2$$

$$= -\frac{1}{4} [1-0] + \frac{1}{4} [2^4-1]$$

$$= -\frac{1}{4} + 4 - \frac{1}{4}$$

$$= 4 - \frac{2}{4}$$

$$= \frac{16-2}{4}$$

$$= \frac{14}{4} = \left(\frac{7}{2} \right)$$

Problem:

17)

$$f(x) = \begin{cases} |x+4| & ; x \leq 0 \\ -x+4 & ; x > 0 \end{cases}$$

(a) $\int_{-8}^{-5} f(x) dx$

$$= \int_{-8}^{-5} (x+4) dx$$

$$= \left[\frac{x^2}{2} + 4x \right]_{-8}^{-5}$$

$$= \frac{25}{2} - 20 - \frac{64}{2} + 32$$

$$= -\frac{15}{2}$$

Area can't be negative. So, $\frac{15}{2}$

[Ans.]

(b) $\int_{-4}^4 f(x) dx$

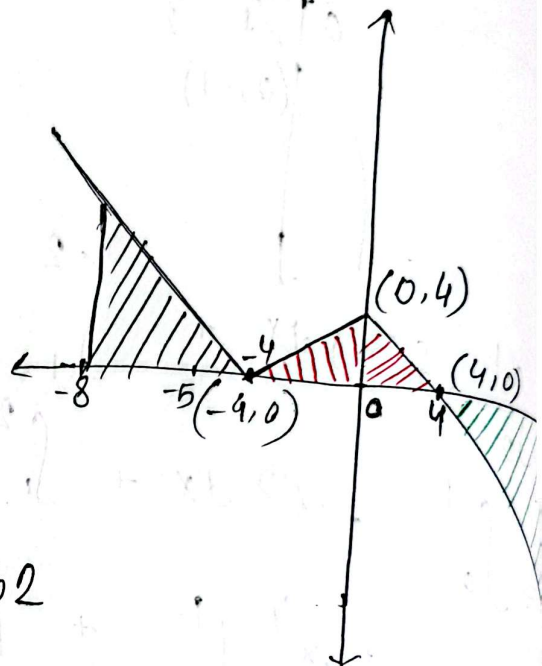
$$= \int_{-4}^0 (x+4) dx + \int_0^4 (-x+4) dx$$

$$= \left[\frac{x^2}{2} + 4x \right]_{-4}^0 + \left[-\frac{x^2}{2} + 4x \right]_0^4$$

$$= -\left(\frac{16}{2} - 16 \right) + \left(-\frac{16}{2} + 16 \right)$$

$$= 16$$

[Ans.]





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(c) $\int_4^6 f(x) dx$

$$= \int_4^6 -(-x+4) dx$$

$$= \int_4^6 x-4 dx$$

$$= \left[\frac{x^2}{2} - 4x \right]_4^6$$

$$= \frac{6^2}{2} - 24 - \frac{16}{2} + 16$$

$$= 2$$

[Ans.]

$$y = x - 4$$

$$y = 0 ; 0 = x - 4$$

$$\Rightarrow 0 + 4 = x$$

$$x = 4$$

Problem: 18)

$$f(x) = \begin{cases} 2x+3 & ; x \leq 0 \\ 3 & ; x > 0 \end{cases}$$

Now, $\int_{-3/2}^2 f(x) dx$

$$= \int_{-3/2}^0 f(x) dx + \int_0^2 f(x) dx$$

$$= \int_{-3/2}^0 (2x+3) dx + \int_0^2 3 dx$$

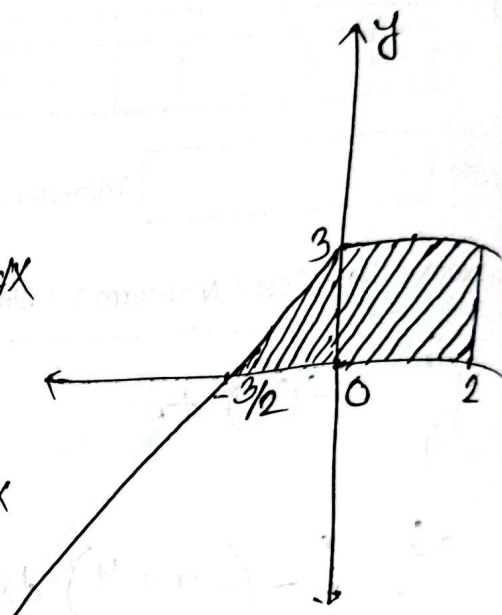
$$= \left[2 \frac{x^2}{2} + 3x \right]_{-3/2}^0 + \left[3x \right]_0^2$$

$$= \left[\left(-\frac{3}{2} \right)^2 - 3 \left(-\frac{3}{2} \right) \right] + 3[2-0]$$

$$= - \left[-\frac{9}{4} \right] + 6$$

$$= \frac{33}{4}$$

[Ans.]



$2x+3$
 $y=0; x=-3/2$
 $x=0; y=3$

Problem: 19) $\int_0^3 (2x + \sqrt{9-x^2}) dx$

Now, $\int 2x dx$

$$= 2 \cdot \frac{x^2}{2}$$

$$= x^2 + C$$

and $\int \sqrt{9-x^2} dx$

Assume, $x = 3 \sin \theta$

$$\Rightarrow dx = 3 \cos \theta d\theta$$

$$\begin{cases} \sqrt{a^2 - x^2} \\ x = a \sin \theta \end{cases}$$

Now, $\int \sqrt{9-9\sin^2 \theta} \cdot 3 \cos \theta d\theta$

$$= \int \sqrt{9(1-\sin^2 \theta)} \cdot 3 \cos \theta d\theta$$

$$= \int 3 \sqrt{\cos^2 \theta} \cdot 3 \cos \theta d\theta$$

$$= \int 9 \cos^2 \theta d\theta$$

$$= 9 \int \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{9}{2} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{9}{2} \int d\theta + \frac{9}{2} \int \cos 2\theta d\theta$$

$$= \frac{9}{2} \theta + \frac{9}{2} \frac{\sin 2\theta}{2}$$

$$= \frac{9}{2} \theta + \frac{9}{4} \sin 2\theta$$

$$= \frac{9}{2} \theta + \frac{9}{4} 2 \sin \theta \cos \theta$$

$$= \frac{9}{2} \theta + \frac{9}{2} \sin \theta \cos \theta$$

$$= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) + \frac{9}{2} \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3}$$

$$= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) + \frac{x}{2} \sqrt{9-x^2} + C$$

Now,

$$\left[x^2 + \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) + \frac{x \sqrt{9-x^2}}{2} \right]_0^3 \cos \theta = \frac{\text{Base}}{\text{hypotenuse}}$$

$$= 9 + \frac{9}{2} \frac{\pi}{2} + 0 -$$

(0)

$$= 9 + \frac{9\pi}{4}$$

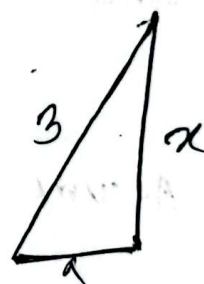
[Ans.]

$$x = 3 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{x}{3}$$

$$\therefore \theta = \sin^{-1} \left(\frac{x}{3} \right)$$

$$\sin = \frac{\text{Opposite}}{\text{hypotenuse}}$$



$$x^2 + a^2 = 9$$

$$a^2 = 9 - x^2$$

$$a = \sqrt{9-x^2}$$

$$\cos \theta = \frac{\text{Base}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\sqrt{9-x^2}}{3}$$



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Problem: 20)

(a) $y^2 = 4x$ and $y = x - 8$

Now,

$$(x-8)^2 = 4x$$

$$\Rightarrow x^2 - 16x + 64 = 4x$$

$$\Rightarrow x^2 - 16x - 4x + 64 = 0$$

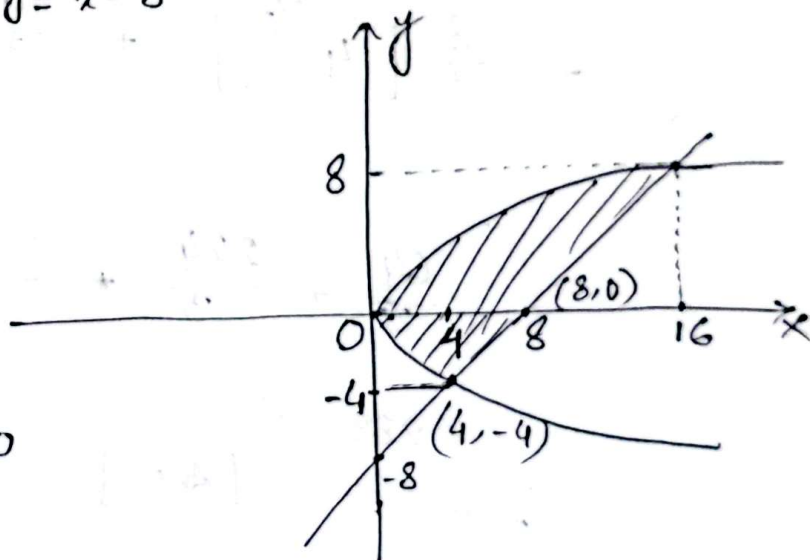
$$\Rightarrow x^2 - 20x + 64 = 0$$

$$x = 16, 4$$

$$x = 16 ; \quad y = 16 - 8 = 8$$

$$x = 4 ; \quad y = 4 - 8 = -4$$

$$y = \sqrt{4x}$$
$$\therefore y = \pm 2\sqrt{x}$$



with respect to x ,

$$\int_0^4 f(x) dx + \int_4^{16} f(x) dx$$

$$= \int_0^4 \{2\sqrt{x} - (-2\sqrt{x})\} dx +$$

$$\int_4^{16} (2\sqrt{x} - x + 8) dx$$

$$= \int_0^4 4\sqrt{x} \, dx + \int_4^{16} 2\sqrt{x} \, dx$$

$$= \int_0^4 4x^{\frac{1}{2}} \, dx + \int_4^{16} 2x^{\frac{1}{2}} \, dx$$

$$= 4 \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4 + 2 \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_4^{16} - 4 \left[\frac{x^2}{2} \right]_4^{16} + [8x]_4^{16}$$

$$= \frac{8}{3} [4^{3/2}] + \frac{4}{3} [16^{3/2} - 4^{3/2}] - 2[16^2 - 4^2] + 8[16 - 4]$$

$$= \frac{64}{3} + \frac{224}{3} - 480 + 96$$

$$= -288 + 72$$

$$= -216 \quad [\text{Ans.}]$$

With respect to y,

$$\int_{-4}^8 f(x) \, dx$$

$$= \int_{-4}^8 (x - 8 - 2\sqrt{x}) \, dx$$

$$= \left[\frac{x^2}{2} - 8x - \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_{-4}^8$$

$$= -62.16 - [$$

$$= \int_{-4}^8 \left(y + 8 - \frac{y^2}{4} \right) dy$$

$$= 72 \text{ unit}^2$$

[Ans.]

$$\int_4^{16}$$

$$16^2 - 4^2$$

$$\int$$

$$\int_{-4}^8$$



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20) (ii) $y = 1+x$ and $y = 2x^2$

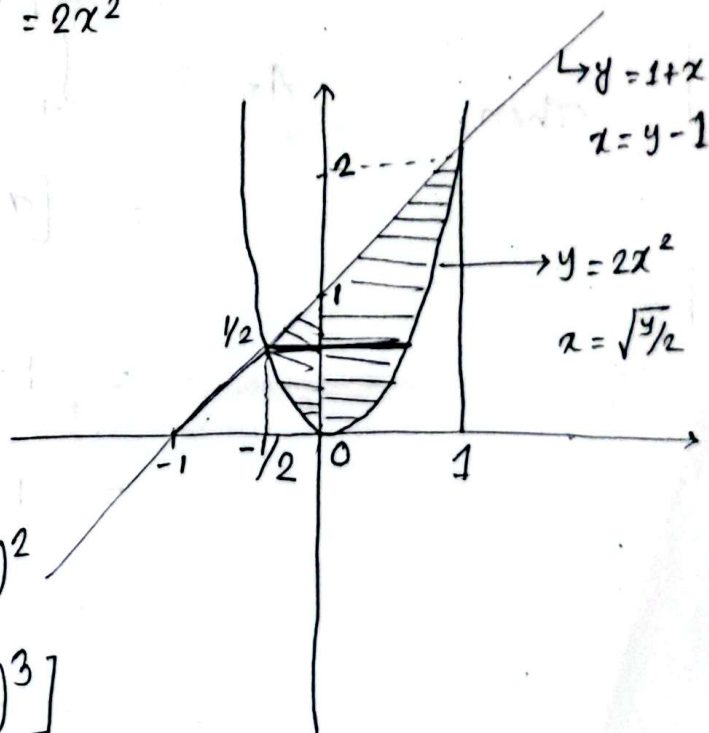
$$A_x = \int_{-1/2}^1 (1+x-2x^2) dx$$

$$= \left[x + \frac{x^2}{2} - 2 \frac{x^3}{3} \right]_{-1/2}^1$$

$$= \left[1 + \frac{1}{2} - \frac{2}{3} \right] - \left[-\frac{1}{2} + \frac{1}{2} \left(-\frac{1}{2} \right)^2 - \frac{2}{3} \left(-\frac{1}{2} \right)^3 \right]$$

$$= \frac{5}{6} + \frac{7}{24}$$

$$\approx 1.125$$



And, $A_y = 2 \int_0^{1/2} \sqrt{y/2} dy + \int_{1/2}^2 \sqrt{\frac{1}{2}y} - (y-1) dy$

$$= \frac{2}{\sqrt{2}} \left[\frac{2}{3} y^{3/2} \right]_0^{1/2} + \left[\frac{1}{\sqrt{2}} \times \frac{2}{3} y^{3/2} - \frac{1}{2} y^2 + y \right]_{1/2}^2$$

$$= 1.125 \text{ [Ans.]}$$

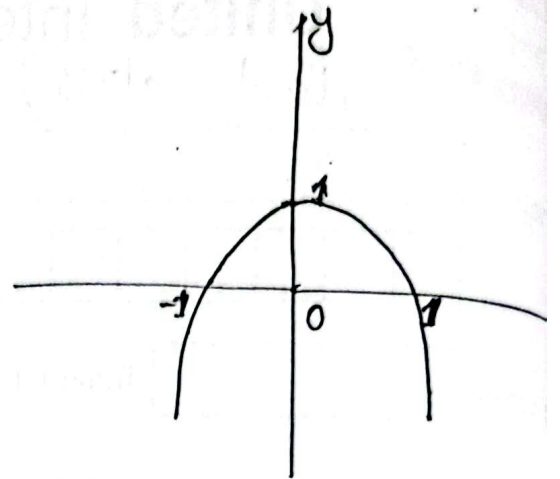
(iv) $y = 1 - x^2$ and $y = 0$

Now, $1 - x^2 = 0$

$$x = \pm 1$$

$$x = 1 ; y = 0$$

$$x = -1 ; y = 0$$



Then, $A_x = \int_{-1}^1 (1 - x^2) dx$

$$= \left[x - \frac{x^3}{3} \right]_{-1}^1$$

$$= 1 - \frac{1}{3} - \left(-1 - \frac{(-1)^3}{3} \right)$$

$$= 1 - \frac{1}{3} + 1 - \frac{1}{3}$$

$$= 2 - \frac{2}{3}$$

$$= \frac{4}{3}$$

and $A_y = 2 \int_0^1 \sqrt{1-y} dy$

$$= 2 \left[-\frac{2}{3} (1-y)^{\frac{3}{2}} \right]_0^1$$

$$= -\frac{4}{3} (1-1)^{\frac{3}{2}} + \frac{4}{3} (1-0)^{\frac{3}{2}}$$

$$= 0 + \frac{4}{3}$$

$$= \frac{4}{3} \quad [\text{Ans.}]$$

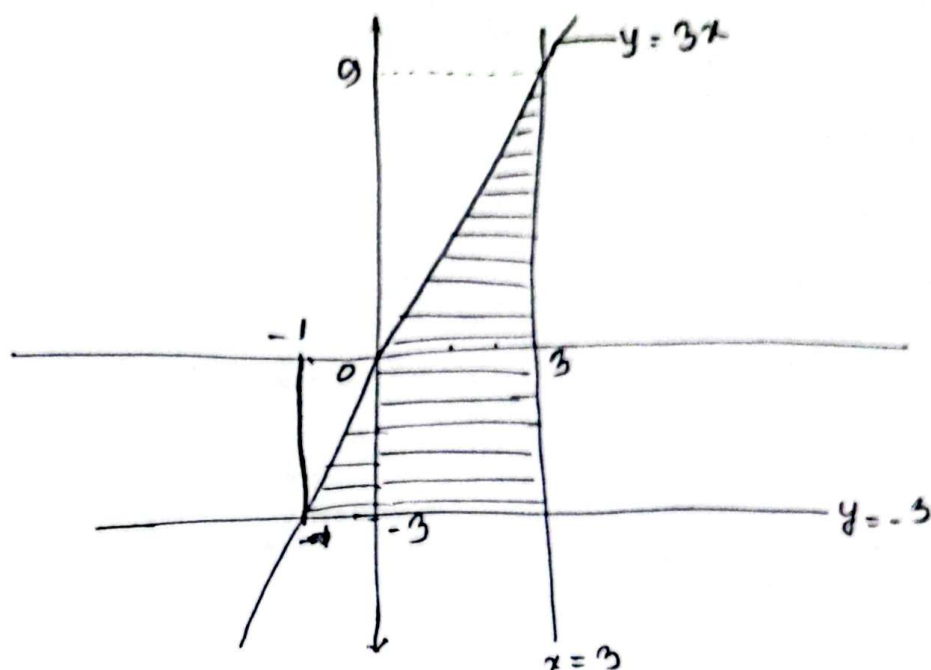
(v) $y = 3x$, $y = -3$ and $x = 3$

$$3x = -3$$

$$x = -1$$

$$x = -1 ; y = 3(-1) = -3$$

$$x = 3 ; y = 9$$



Now, $Ax = \int_{-1}^3 (3x + 3) dx$

$$= \left[3 \frac{x^2}{2} + 3x \right]_{-1}^3$$

$$= \frac{3}{2} (3)^2 + 9 - \left[\frac{3}{2} (-1)^2 + 3(-1) \right]$$

$$= 24$$

and $Ay = \int_{-3}^9 \left(3 - \frac{1}{3}y \right) dy$

$$= 3[y]_{-3}^9 - \frac{1}{3} \left[\frac{y^2}{2} \right]_{-3}^9$$

$$= 3[9 + 3] - \frac{1}{6} [9^2 - (-3)^2]$$

$$= 36 - 12$$

$$= 24 \quad [\text{Ans.}]$$



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Problem: 21) a) Area, $A = \int_0^3 \{10 - y^2 - (y-2)^2\} dy$

$$= \int_0^3 (10 - y^2 - y^2 + 4y - 4) dy$$

$$= \left[10y - \frac{y^3}{3} - \frac{y^3}{3} + 4 \frac{y^2}{2} - 4y \right]_0^3$$

$$= 10 \times 3 - \frac{1}{3}(3)^3 - \frac{1}{3}(3)^3 + 2(3)^2 - 12$$

$$= 30 - \frac{27}{3} - \frac{27}{3} + 18 - 12$$

$$= \frac{90 - 27 - 27 + 54 - 36}{3}$$

$$= \frac{54}{3}$$

$$= 18$$

[Ans]

$$\begin{aligned}
 \underline{\underline{b)}} \quad & \int_{-1}^0 3(x^3 - x) dx + \int_0^1 -3(x^3 - x) dx \\
 &= 3 \int_{-1}^0 (x^3 - x) dx + 3 \int_0^1 (x^3 - x) dx \\
 &= 3 \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + 3 \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 \\
 &= 3 \left[- \left(\frac{(-1)^4}{4} - \frac{(-1)^2}{2} \right) \right] + 3 \left[\frac{(1)^4}{4} - \frac{1^2}{2} \right] \\
 &= 3 \left[-\frac{1}{4} + \frac{1}{2} \right] + 3 \left[\frac{1}{4} - \frac{1}{2} \right] \\
 &= 3 \left[\frac{2-1}{4} \right] + 3 \left[\frac{1-2}{4} \right] \\
 &= \frac{3}{4} - \frac{3}{4} \\
 &= 0 \quad [\text{Ans.}]
 \end{aligned}$$

$$\begin{aligned}
 \underline{\underline{c)}} \quad & \int_{-1}^2 \{x^2 - 2x + 2 - (x^2 + 6)\} dx \\
 &= \int_{-1}^2 (x^2 - 2x + 2 + x^2 - 6) dx \\
 &= 2 \left[\frac{x^3}{3} \right]_{-1}^2 - 2 \left[\frac{x^2}{2} \right]_{-1}^2 + 4 \left[x \right]_{-1}^2 \\
 &= \frac{2}{3} [2^3 - (-1)^3] - [2^2 - (-1)^2] + 4 [2 - (-1)]
 \end{aligned}$$

$$= \frac{2}{3} [8+1] - [4-1] - 4 [2+1]$$

$$= \frac{2}{3} \times 9 - 3 - 12$$

$$= 6 - 3 - 12$$

$$= 6 - 15$$

$$= -9$$

$$\text{so, Area} = 9 \quad [\text{Ans.}]$$

$$d) \int_{-2}^2 (y+1 - \frac{1}{2}y^2 + 3) dy$$

$$= \left[\frac{y^2}{2} + y - \frac{1}{2} \frac{y^3}{3} + 3y \right]_{-2}^2$$

$$= \left[\frac{2^2}{2} + 2 - \frac{1}{6} 2^3 + 6 \right] - \left[\frac{1}{2} (-2)^2 - 2 - \frac{1}{6} (-2)^3 - 6 \right]$$

$$= \left[2 + 2 - \frac{8}{6} + 6 \right] - \left[2 - 2 + \frac{8}{6} - 6 \right]$$

$$= \left[\frac{12 + 12 - 8 + 36}{6} \right] - \left[\frac{12 - 12 + 8 - 36}{6} \right]$$

$$= \frac{52}{6} + \frac{28}{6}$$

$$=$$

$$= \frac{80}{6}$$

Area, A

$$= \int_{-1}^0 (x - 5x + x^2) dx + \int_0^5 (5x - x^2 - x) dx$$

$$= \left[\frac{x^2}{2} - \frac{5x^2}{2} + \frac{x^3}{3} \right]_{-1}^0 + \left[\frac{5x^2}{2} - \frac{x^3}{3} - \frac{x^2}{2} \right]_0^5$$

$$= - \left[\frac{(-1)^2}{2} - \frac{5(-1)^2}{2} + \frac{1}{3}(-1)^3 \right] + \left[\frac{5}{2}(5)^2 - \frac{1}{3}(5)^3 - \frac{1}{2}(5)^2 \right]$$

$$= - \left[\frac{1}{2} - \frac{5}{2} + \frac{1}{3} \right] + \left[\frac{5}{2} \times 25 - \frac{1}{3} \times 125 - \frac{1}{2} \times 25 \right]$$

$$= - \frac{1}{2} + \frac{5}{2} + \frac{1}{3} + \frac{125}{2} - \frac{125}{3} - \frac{25}{2}$$

$$= \frac{-3 + 15 + 2 + 375 - 250 - 75}{6}$$

$$= \frac{64}{6}$$

$$= \frac{32}{3}$$

[Ans.]



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f) Area, A

$$= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$
$$= [-\sin x]_0^{\pi/4} - [\cos x]_0^{\pi/4} + [\cos x]_{\pi/4}^{\pi/2} + [\sin x]_{\pi/4}^{\pi/2}$$
$$= [-\sin \pi/4 + 0] - [\cos \pi/4 - \cos 0] + [\cos \pi/2 - \cos \pi/4] + [\sin \pi/2 - \sin \pi/4]$$
$$= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1 + 0 + \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}}$$

(Do the calculation)

$$= \frac{2}{\sqrt{2}}$$
$$= \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

[Ans.]

g) Area, A

$$= \int_{-1}^0 (x^2 - \sqrt{x}) dx + \int_0^1 (\sqrt{x} - x^2) dx$$

$$= \left[\frac{x^3}{3} - \frac{x^{4/3}}{4/3} \right]_{-1}^0 + \left[\frac{x^{4/3}}{4/3} - \frac{x^3}{3} \right]_0^1$$

$$= - \left[\frac{(-1)^3}{3} - \frac{3(-1)^{4/3}}{4} \right] + \left[\frac{3}{4} - \frac{1}{3} \right]$$

$$= - \left[-\frac{1}{3} + \frac{3}{4} \right] + \left[\frac{9-4}{12} \right]$$

$$= -\frac{3}{4} + \frac{1}{3} + \frac{5}{12}$$

$$= \frac{-9 + 4 + 5}{12}$$

$$= \frac{0}{12} = 0$$

So, Area, A = $\frac{5}{6}$ [Ans.]

h) $\int_4^{5.5} \left(9 - \left(\frac{x}{2} \right)^2 - 6 + x \right) dx$

$$= \left[9x - \frac{1}{4} \frac{x^3}{3} - 6x + \frac{x^2}{2} \right]_4^{5.5}$$

$\Rightarrow \square$

[Ans.]

em: 23)

a) $\int_0^2 |3x-6| dx$

$= \int_0^2 6-3x dx$

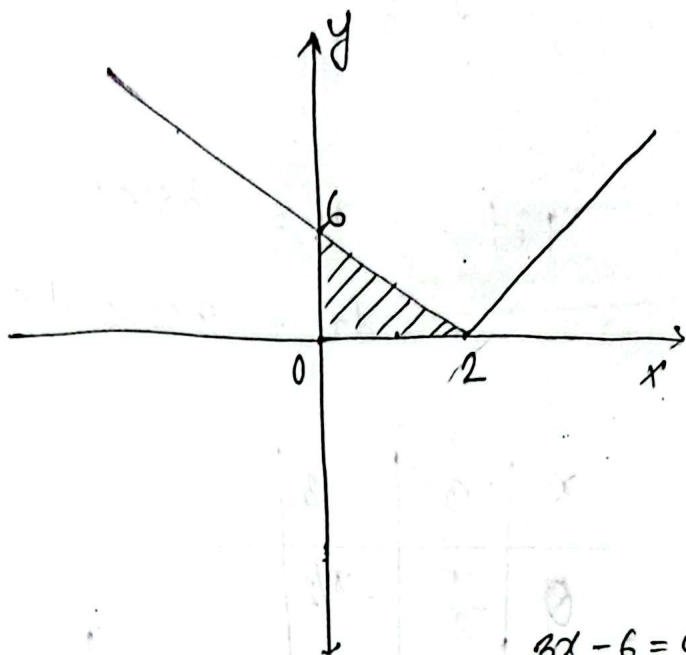
$= [6x]_0^2 - 3\left[\frac{x^2}{2}\right]_0^2$

$= 6 \times 2 - \frac{3}{2} \times 2^2$

$= 12 - 6$

$= 6$

[Ans.]



$3x-6=0$

$y=0; 3x=6$

$x=2$

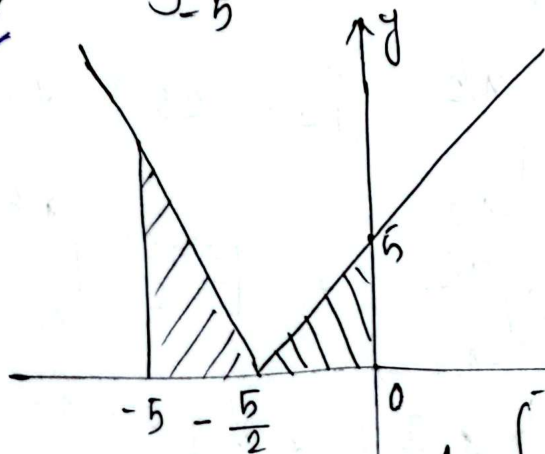
$x=0; y=-6$

$-3x+6=y$

$x=0; y=6$

$y=0; -3x=-6$
 $x=2$

b) $\int_{-5}^0 |2x+5| dx$



$A = \int_{-5}^{-5/2} f(x) dx + \int_{-5/2}^0 f(x) dx$

$y = -(2x+5)$
 $x=0; y=-5$
 $-2x-5=0$
 $-2x=5$
 $x=-5/2; y=0$

$\int_{-5}^0 2x+5 dx$

$2\left[\frac{x^2}{2}\right]_{-5}^0 + 5\left[x\right]_{-5}^0$

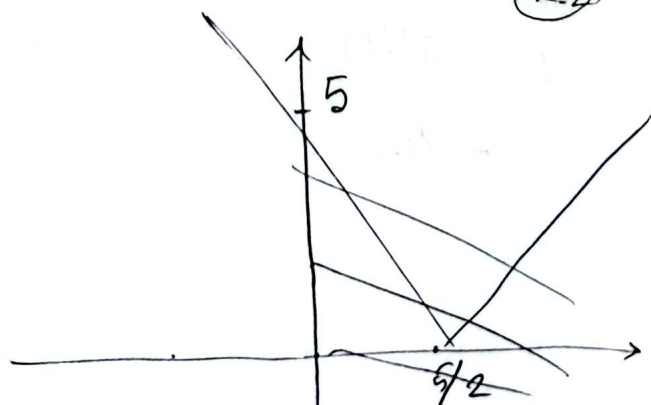
$= \int_{-5}^{-5/2} -(2x+5) dx + \int_{-5/2}^0 (2x+5) dx$

$y = 2x+5$

$x=0; y=5$

$y=0; 2x=-5$

$x=-5/2$



$= 2 \times \frac{1}{2} [0 - (-5)^2] + 5(0+5)$

c) $\int_{-3}^3 \sqrt{9-x^2} dx$

Let, $x = 3 \sin \theta$

$dx = 3 \cos \theta d\theta$

x	3	-3
θ	$\frac{\pi}{2}$	$-\frac{\pi}{2}$

$\sin \theta = 3$

$\sin \theta = 1$

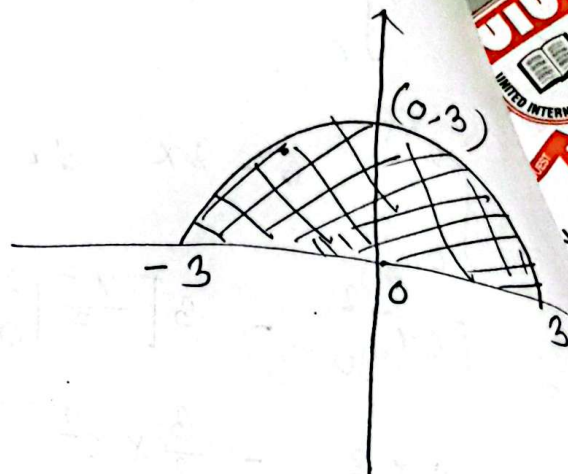
$\theta = \sin^{-1}(1)$

$= \frac{\pi}{2}$

$\sin \theta = -1$

$\theta = \sin^{-1}(-1)$

$= -\frac{\pi}{2}$



Now, $\int_{-\pi/2}^{\pi/2} \sqrt{9-9\sin^2 \theta} \cdot 3 \cos \theta d\theta$

$= \int_{-\pi/2}^{\pi/2} 3 \cos \theta \cdot 3 \cos \theta d\theta$

$= \int_{-\pi/2}^{\pi/2} 9 \cos^2 \theta d\theta$

$= \frac{9}{2} \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta$

$= \frac{9}{2} \left[\theta \right]_{-\pi/2}^{\pi/2} + \left[\frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$

$= \frac{9}{2} \left[\frac{\pi}{2} + \frac{\pi}{2} \right] + \frac{1}{2} [\sin \pi - \sin(-\pi)]$

$= \frac{9}{2} \times \frac{2\pi}{2} + \left(\frac{1}{2} \times 0 \right)$

$= \frac{9\pi}{2}$

[Ans.]



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Problem: (f)
(24) $\int \frac{\sin(1/x)}{3x^2} dx$

Let, $1/x = u$

$$\Rightarrow -x^{-2} dx = du$$

$$\Rightarrow \frac{dx}{x^2} = -du$$

Then, $-\int \sin u du$

$$= \cos u + C$$

$$= \cos(1/x) + C \quad [\text{Ans.}]$$

(b) $\int x^{1/3} (2-x)^2 dx$

$$= \int x^{1/3} (4 - 4x + x^2) dx$$

$$= \int (4x^{1/3} - 4x^{4/3} + x^{7/3}) dx$$

$$= \frac{3 \cdot 4}{4} x^{4/3} - \frac{12}{4} x^{7/3} + \frac{3}{10} x^{10/3}$$

[Ans.]

$$\begin{aligned} & \frac{1}{3} + 1 \\ &= \frac{1+3}{3} \\ &= \frac{4}{3} \\ & \frac{2+1}{3} \\ &= \frac{6+1}{3} \end{aligned}$$

$$(c) \int \frac{\sin x}{\cos^2 x} dx$$

$$\text{Let, } \cos x = u$$

$$\Rightarrow -\sin x dx = du$$

$$\therefore \sin x dx = -du$$

$$\text{Then, } -\int \frac{1}{u^2} du$$

$$= -\frac{u^{-2+1}}{-2+1} + C$$

$$= u^{-1} + C$$

$$= (\cos x)^{-1} + C$$

$$= \frac{1}{\cos x} + C$$

$$= \sec x + C \quad [\text{Ans.}]$$

$$(e) \int \frac{x}{\sqrt{4-5x^2}} dx$$

$$\text{Let, } 4-5x^2 = u$$

$$\Rightarrow -10x dx = du$$

$$\therefore x dx = -\frac{1}{10} du$$

$$\int \frac{-\frac{1}{10} du}{\sqrt{u}}$$

$$= -\frac{1}{10} \int \frac{du}{\sqrt{u}}$$

$$= -\frac{1}{10} \frac{u^{\frac{3}{2}}}{\frac{3}{2}}$$

$$= -\frac{1}{15} u^{\frac{3}{2}} + C$$

$$= -\frac{1}{15} (4-5x^2)^{\frac{3}{2}} + C$$

$$[\text{Ans.}]$$

$$(1) \int \sec^6 x \tan^6 x dx$$

$$= \int (\sec x \tan x)^6 dx$$

$$= \int \sec^5 x \tan^5 x \cdot \sec x \tan x dx$$

$$\text{Let, } \sec x = u$$

$$\Rightarrow \sec x \tan x dx = du$$

$$\begin{aligned}
 & \int \sec^6 x \tan^6 x \, dx \\
 &= \int (\sec^2 x)^3 (\tan^2 x)^3 \, dx \\
 &= \int (\sec^2 x)^3 (\sec^2 x - 1)^3 \, dx \\
 &\text{Let,}
 \end{aligned}$$

$$\begin{aligned}
 & \underline{\underline{(1)}} \quad \int \sec^6 x \tan^6 x \, dx \\
 &= \int \tan^6 x \cdot \sec^2 x \cdot (1 + \tan^2 x)^2 \, dx
 \end{aligned}$$

$$\text{Let, } u = \tan x$$

$$\therefore du = \sec^2 x \, dx$$

$$\text{Then, } \int u^6 (1 + u^2)^2 \, du$$

$$= \int u^6 (1 + 2u^2 + u^4)^2 \, du$$

$$= \int (u^6 + 2u^8 + u^{10}) \, du$$

$$= \int u^6 \, du + 2 \int u^8 \, du + \int u^{10} \, du$$

$$= \frac{u^7}{7} + 2 \frac{u^9}{9} + \frac{u^{11}}{11} + C$$

$$= \frac{1}{7} \tan^7 x + \frac{2}{9} \tan^9 x + \frac{1}{11} \tan^{11} x + C$$

[Ans]

Problem: 26) (a) $\int x^2 \ln x \, dx$

$$= \ln x \int x^2 \, dx - \int \left\{ \frac{d}{dx} (\ln x) \int x^2 \, dx \right\} dx$$

$$= \ln x \frac{x^3}{3} - \int \frac{1}{x} \frac{x^3}{3} \, dx$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 \, dx$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{3} \frac{x^3}{3} + C$$

$$= \frac{1}{3} x^3 \ln x - \frac{x^3}{9} + C \quad [\text{Ans.}]$$

(c) $\int x^2 \cos x \, dx$

$$= x^2 \int \cos x \, dx - \int \left\{ \frac{d}{dx} (x^2) \int \cos x \, dx \right\} dx$$

$$= x^2 \sin x - \int 2x \sin x \, dx$$

$$= x^2 \sin x - 2 \int x \sin x \, dx$$

$$= x^2 \sin x - \left[x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (x) \int \sin x \, dx \right\} dx \right]$$

$$= x^2 \sin x - \left[-x \cos x - \int -\cos x \, dx \right]$$

$$= x^2 \sin x + x \cos x - \sin x + C$$

[Ans.]



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$$(L) \int e^{-2x} \sin 3x \, dx$$

$$= \sin 3x \int e^{-2x} \, dx - \int \left\{ \frac{d}{dx} (\sin 3x) \int e^{-2x} \, dx \right\} dx$$

$$= \sin 3x \frac{e^{-2x}}{-2} - \int 3 \cos 3x \cdot \frac{e^{-2x}}{-2} \, dx$$

$$= -\frac{1}{2} e^{-2x} \sin(3x) + \frac{3}{2} \int e^{-2x} \cos 3x \, dx$$

$$= -\frac{1}{2} e^{-2x} \sin(3x) + \frac{3}{2} \left[\cos 3x \int e^{-2x} \, dx - \int \left\{ \frac{d}{dx} (\cos 3x) \int e^{-2x} \, dx \right\} dx \right]$$

$$= -\frac{1}{2} e^{-2x} \sin(3x) + \frac{3}{2} \left[\cos 3x \cdot \frac{e^{-2x}}{-2} - \int -3 \sin 3x \frac{e^{-2x}}{-2} \, dx \right]$$

$$= -\frac{1}{2} e^{-2x} \sin(3x) + \frac{3}{2} \left[-\frac{1}{2} \cos(3x) e^{-2x} - \frac{3}{2} \int e^{-2x} \, dx \right]$$

$$= -\frac{1}{2} e^{-2x} \sin(3x) - \frac{3}{4} \cos(3x) e^{-2x}$$

$$\therefore \int = \frac{2}{5} \left(-\frac{1}{2} e^{-2x} \sin(3x) - \frac{3}{4} \cos(3x) e^{-2x} \right) + C$$

(b) $\int_0^{0.3} \frac{x}{(9-25x^2)^{3/2}} dx$

Let, $9-25x^2 = u$

$\Rightarrow -50x dx = du$

$\therefore x dx = \frac{du}{-50}$

Then, $\int \frac{\frac{du}{-50}}{u^{3/2}}$

$= -\frac{1}{50} \int u^{-3/2} du$

$= -\frac{1}{50} \frac{u^{-1/2}}{-1/2} + C$

$= -\frac{1}{25} (9-25x^2)^{-1/2} + C$

[Ans.]

(g) $\int \frac{\sqrt{x^2-9}}{x^3} dx$

Let, $u = \sqrt{x^2-9}$

$\Rightarrow du = \frac{1}{2\sqrt{x^2-9}} \cdot 2x$

$\therefore du = \frac{x}{\sqrt{x^2-9}} dx$

$\Rightarrow u^2 = x^2-9$
 $\therefore x^2 = u^2+9$



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(8) $\int \frac{\sqrt{x^2 - 9}}{x^3} dx$

$$\sqrt{x^2 - a^2}$$

$$\text{Let, } x = a \sec \theta$$

$$\text{Let, } x = 3 \sec \theta$$

$$dx = 3 \sec \theta \tan \theta d\theta$$

$$\text{Then, } I = \int \frac{3 \sqrt{\sec^2 \theta - 1}}{27 \sec^3 \theta} 3 \sec \theta \tan \theta d\theta$$

$$= \int \frac{3 \tan \theta}{27 \sec^2 \theta} 3 \tan \theta d\theta$$

$$\frac{\cancel{\sin^2 \theta}}{\cancel{\cos^2 \theta}} \times \cos^2 \theta = \frac{1}{3} \int \tan^2 \theta \cos^2 \theta d\theta$$

$$\sin^2 \theta = 1 - \cos 2\theta \quad \left] \quad = \frac{1}{3} \int \sin^2 \theta d\theta$$

$$= \frac{1}{3} \times \frac{1}{2} \int 2 \sin^2 \theta d\theta$$

$$= \frac{1}{6} \int (1 - \cos 2\theta) d\theta$$

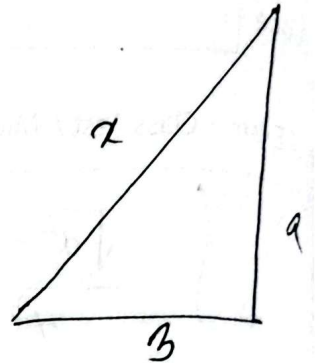
$$= \frac{1}{6} \left[0 - \frac{\sin 2\theta}{2} \right] + C$$

$$= \frac{1}{6} [0 - \sin \theta \cos \theta] + C$$

since, $\sec \theta = \frac{x}{3}$

$$\cos \theta = \frac{3}{x}$$

$$\sin \theta = \frac{\sqrt{x^2 - 9}}{x}$$



$$3^2 + a^2 = x^2$$

$$a^2 = x^2 - 9$$

$$a = \sqrt{x^2 - 9}$$

$$\int \frac{\sqrt{x^2 - 9}}{x^3} dx$$

$$= \frac{1}{6} \left[\sec^{-1} \left(\frac{x}{3} \right) - \frac{\sqrt{x^2 - 9}}{x} \cdot \frac{3}{x} \right] + C$$

[Ans.]

$$\begin{cases} \cos \theta = \frac{\text{base}}{\text{hypo}} \\ \sin \theta = \frac{\text{Perpendic}}{\text{hypo}} \end{cases}$$

(i) $\int \frac{dx}{x^2 \sqrt{x^2 + 25}}$

Let, $x = 5 \tan u \quad \therefore u = \tan^{-1} \left(\frac{x}{5} \right)$

$$dx = 5 \sec^2 u du$$

$$\int \frac{5 \sec^2 u}{25 \tan^2 u \sqrt{25 \tan^2 u + 25}} du$$

$$= \int \frac{\sec^2 u}{\tan^2 u \sqrt{25 \tan^2 u + 25}} du$$

$$= \int \frac{\sec u}{\tan^2 u} du$$

$$= \int \frac{\cos u}{\sin^2 u} du$$

$$t, \quad v = \sin u$$

$$dv = \cos u du$$

$$\text{hence, } \int v^{-2} dv$$

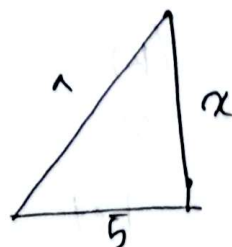
$$= \frac{v^{-2+1}}{-2+1} + C$$

$$= -v^{-1} + C$$

$$= -\sin^{-1} u + C$$

$$= -\sin^{-1} \left(\tan^{-1} \frac{x}{5} \right) + C$$

$$= -\sin^{-1} \left(\sin^{-1} \frac{x}{\sqrt{x^2+25}} \right) + C = -\frac{x}{\sqrt{x^2+25}} + C \quad [\text{Ar}]$$



$$x^2 + 25 = a^2$$

$$a = \sqrt{x^2 + 25}$$

$$(j) \int \frac{dx}{\sqrt{x^2 + 4x + 5}}$$

$$= \int \frac{dx}{\sqrt{(x+2)^2 + 1}}$$

$$\text{Let, } x+2 = u$$

$$dx = du$$

$$\text{Thus, } \int \frac{du}{\sqrt{u^2 + 1}}$$

$$= \ln(u + \sqrt{u^2 + 1}) + C$$

$$= \ln(x+2 + \sqrt{(x+2)^2 + 1}) + C$$

[Ans.]

$$(k) \int \frac{\sqrt{x-3}}{x} dx$$

$$\text{Let, } u = \sqrt{x-3}$$

$$\Rightarrow du = \frac{dx}{2\sqrt{x-3}}$$

$$\therefore dx = 2\sqrt{x-3} du$$

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Then,
$$2 \int \frac{u^2 du}{u^2 + 3}$$

$$= \int \frac{u^2 + 3 - 3}{u^2 + 3} du$$

$$= 2 \int \left(1 - \frac{3}{u^2 + 3} \right) du$$

$$= 2u - 6 \tan^{-1} \frac{u}{\sqrt{3}}$$

$$= \sqrt{x-3} - 6 \tan^{-1} \frac{\sqrt{x-3}}{\sqrt{3}} + C$$

[Ans.]

$$\int \int \frac{\sqrt{x-3}}{2\sqrt{x-3}} \cdot 2\sqrt{x-3}$$

$2 \int u^2$

$$(1) \int \frac{dx}{2x^2 + 4x + 7}$$

$$= \int \frac{dx}{(\sqrt{2}x + \sqrt{2})^2 + 5}$$

$$\text{Let, } u = \frac{\sqrt{2}(x+1)}{\sqrt{5}}$$

$$\frac{du}{dx} = \frac{\sqrt{2}}{\sqrt{5}}$$

$$\therefore dx = \frac{\sqrt{5}}{\sqrt{2}} du$$

$$\text{Then, } \int \frac{\sqrt{5}}{\sqrt{2}(u^2 + 1)} du$$

$$= \frac{1}{\sqrt{2}\sqrt{5}} \int \frac{1}{u^2 + 1} du$$

$$= \frac{1}{\sqrt{2}\sqrt{5}} \tan^{-1} u + C$$

$$= \frac{1}{\sqrt{10}} \tan^{-1} \left[\frac{\sqrt{2}(x+1)}{\sqrt{5}} \right] + C$$

[Ans.]

$$\int \frac{x}{x^2 + 2x + 2} dx$$

$$= \int \left(\frac{2x + 2}{2(x^2 + 2x + 2)} - \frac{1}{x^2 + 2x + 2} \right) dx$$

$$= \int \frac{x+1}{x^2 + 2x + 2} dx - \int \frac{1}{x^2 + 2x + 2} dx$$

Let, $u = x^2 + 2x + 2$

$$du = (2x + 2) dx$$

$$\therefore dx = \frac{du}{2x+2}$$

$$= \int \frac{x+1}{x^2 + 2x + 2} \frac{du}{2(x+1)} - \int \frac{1}{x^2 + 2x + 2} \frac{du}{2x+2}$$

$$= \int \frac{du}{2u} - \int \frac{dx}{x^2 + 2x + 2}$$

$$= \frac{1}{2} \ln u - \int \frac{dx}{(x+1)^2 + 1}$$

$$= \frac{1}{2} \ln u - \int \frac{du}{u^2 + 1} \left[\begin{array}{l} x+1=u \\ dx=du \end{array} \right]$$

$$= \frac{1}{2} \ln u - \tan^{-1} u + C$$

$$= \frac{1}{2} \ln(x^2 + 2x + 2) - \tan^{-1}(x^2 + 2x + 2) + C$$

$$(11) \int \frac{x^5 - 3x^2 + 2}{x^3 - x} dx$$

$$= \int \left(\frac{-3x^2 + x + 2}{x^3 - x} + x^2 + 1 \right) dx$$

$$= - \int \frac{3x^2 - x - 2}{x^3 - x} dx + \int x^2 dx + \int dx$$

$$\text{Thus, } \int \frac{3x^2 - x - 2}{x^3 - x} dx$$

$$= \int \frac{3x^2 - x - 2}{x(x-1)(x+1)} dx$$

$$= \int \left(\frac{1}{x+1} + \frac{2}{x} \right) dx$$

$$= \ln(x+1) + 2 \ln x$$

$$\text{Thus, } I = -\ln(x+1) - 2 \ln x + \frac{x^3}{3} + x + C$$

[Ans.]

$$\frac{3x^2 - x - 2}{x(x-1)(x+1)} = \frac{(x-1)(3x+2)}{x(x-1)(x+1)}$$

$$\frac{3x^2 - 3x + 2x - 2}{3x(x-1) + 2(x-1)}$$

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$$\frac{(x-1)(3x+2)}{x(x-1)(x+1)} =$$

$$\frac{3x+2}{x(x+1)}$$

$$= \frac{A}{x} + \frac{Bx+C}{x(x+1)}$$

$$3x+2 = A(x+1) + Bx+C$$

$$-3+2 = -B+C$$

$$-B+C = -1$$

$$x=0 ; 2 = A+C$$

$$\frac{3x+2}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$

$$3x+2 = A(x+1) + Bx$$

$$x=-1 ; -1 = B(-1) \therefore B=1$$

$$x=0 ; 2=A$$

$$\text{So, } \frac{3x+2}{x(x+1)} = \frac{2}{x} + \frac{1}{x+1}$$

Ch-7.4

$$1) \sqrt{a^2 - x^2}$$

$$; \quad x = a \sin \theta$$

$$2) \sqrt{a^2 + x^2}$$

$$; \quad x = a \tan \theta$$

$$3) \sqrt{x^2 - a^2}$$

$$; \quad x = a \sec \theta$$

Example :

1, 2, 3, ^{*}5, ^{*}6